

“Same Click, Different Risks”: Geography as a Hidden Factor in Web Privacy and Security

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Abstract

Web applications underpin essential global and regional services, yet their S&P guarantees may differ across regions due to varying deployment practices, or even regulatory or development norms. Unfortunately, prior web measurements often failed to capture these and only provide a shallow view of the prevalence of S&P controls and from limited access points.

To address this gap, we conduct a comprehensive study on the top 10K websites of five different countries with distinct data protection regimes and investigate how websites deploy effective client-side S&P controls. Our evaluation spans critical factors, including exposure to tracking, browser fingerprinting, deployment of security-related HTTP headers, and third-party script integrations.

Our findings reveal significant variation in users’ exposure to S&P risks across dimensions. For instance, third-party tracker prevalence peaks in different regions depending on site category: the UAE for globally popular sites, but Germany for regionally popular ones. More broadly, while both global and local websites exhibit deep third-party script inclusion chains, their prevalence, utility and underlying mechanisms diverge significantly by category and geography. Similarly, websites across URL catalogs show comparable adoption rates of security-related HTTP headers and related primitives, but the effective policy configurations and enforcement diverge sharply. We present several case studies illustrating these diverging trends between global and regional websites, discuss potential underlying causes, and provide recommendations for more representative web measurements going forward.

1 Introduction

The web has evolved to become increasingly *accessible*, *dynamic*, and *localized*, aiming to attract users, deliver diverse content across devices, and adapt to individual needs through personalized services. At the same time, this evolution raises significant security and privacy (S&P) concerns for users.

The research community has extensively examined the security and privacy posture of web applications and related

policies [1, 2, 17, 23, 25, 30, 39, 51, 52, 56, 60, 62, 69, 74, 82, 84, 94]. Parallely, researchers also explored users’ awareness, perceptions, and behavioral responses to security and privacy challenges on the web [12, 22, 32, 40, 71, 95]. A typical, yet critical, limitation across these studies is their predominant focus on “Top N” global websites, resulting in a skewed representation of websites and user experiences from regions commonly used as measurement vantage points (e.g., North America and Western Europe). This narrow lens contrasts sharply with the demographic reality where a substantial portion of Internet users are physically located outside these overrepresented regions [42]. Moreover, regions differ widely with their data regulations and privacy-enforcement frameworks, which may cause websites to tailor their development norms and treat their *local* users uniquely. Consequently, the collective findings of these works only paint a partial picture of the *actual* state of S&P across the *World Wide Web*.

More recently, researchers focused on analyzing regional websites, focusing on specific categories, such as social media content, government portals, and news outlets, to evaluate their development, accessibility, and deployment practices (e.g., server-side randomness, tracking mechanisms, or censorship patterns) [9, 10, 11, 31, 68, 81, 83, 88]. While these efforts are important steps toward a holistic understanding of the actual web, they also suffer from three fundamental shortcomings – *representation*, *behavior*, and *comprehensiveness*. First, the choice of URLs to measure S&P controls across regions is crucial to obtain a broader view of the global S&P deployments. Previous studies focused primarily on “Top N” global websites and did not shed light on how S&P attributes are *representative* across regional and global applications. Secondly, most studies relied on data collected via VPNs or cloud-based services (e.g., Amazon S3, Azure). As we later demonstrate, and noted by Böttger et al. [13], such services are frequently server-side fingerprinted by websites, resulting in capturing *behavior* that deviate markedly from the users’ reality. Finally, prior regional measurements focused on a limited set of indicators (e.g., presence of cookie banners or privacy policies) which does not provide a *comprehensive*

view of how client-side S&P signals vary across regions.

In this study, we aim to address the three fundamental challenges above by proposing a robust framework to conduct representative S&P measurements across different countries. We utilize this framework to analyze the client-side security and privacy deployments of both global and regional websites through accessing them under local users’ conditions in *Algeria, Germany, India, the UAE, and the USA*. Our findings reveal substantial cross-country variation in users’ exposure to S&P risks. For instance, even when accessing identical websites, users in different locations encounter different tracking behaviors. Similarly, identical third-party scripts may also exhibit divergent runtime behavior as per users’ access point.

To summarize, our main contributions are as follows:

- We propose a framework to construct a dataset of websites for S&P measurements that captures both global and regional web experiences and are representative of the *World Wide web*.
- We present a large-scale, multi-vantage-point measurement study of S&P behaviors across 5 countries, through physical machines rather than relying on VPN or cloud-based vantage points.
- We systematically measure and compare multiple client-side S&P controls, including third-party tracking, browser fingerprinting, security-related header deployment and script inclusion patterns.
- Through this, we find that both website type and geographic vantage point are important factors shaping websites’ security and privacy postures. Based on these findings, we provide recommendations to the research community and industry on conducting more representative web measurements in the future.

2 Background & Related Work

We now describe various S&P aspects associated with web applications that we measure here. We also outline closely-related prior studies that reported regional S&P practices.

2.1 Privacy on the Web

Stateful Tracking: Stateful tracking refers to tracking techniques that rely on browser-managed storage mechanisms, such as cookies and local storage, to maintain state across page visits or browsing sessions. These mechanisms can be used by first-party websites or embedded third-party resources to support functionality, analytics, advertising, and cross-site tracking. Early studies measured the prevalence of stateful tracking at scale by analyzing browser storage and developing heuristics to distinguish tracking-related values from benign website state, using properties such as value structure, persistence, and entropy [1, 30, 48, 65]. Further, Munir et al. [60] proposed machine-learning techniques to detect

first-party cookies related to tracking. Orthogonally, several researchers used curated third-party tracking blocklists, such as EasyList [26] and EasyPrivacy [28], to measure the prevalence of third-party tracking in the wild [49, 54, 92].

Browser Fingerprinting: Browser fingerprinting is a tracking technique that leverages built-in JavaScript APIs to collect and combine device- and browser-specific attributes into a high-entropy identifier used to recognize browsers, and subsequently users, across visits and websites [14, 29]. Prior work showed that a wide range of browser APIs contribute to fingerprinting, and categorized these techniques solely based on the signals they exploit. Common browser fingerprinting techniques include canvas-based fingerprinting, which derives identifiers from the list of available fonts and image rendering behavior; AudioContext fingerprinting, which exploits variations in audio signal processing; and WebRTC-based fingerprinting, which leverages network-related information such as local and public IP addresses [1, 29, 30, 39, 58]. Prior work has measured browser fingerprinting in the wild. Acar et al. [1] found canvas fingerprinting on over 5% of the top 100K websites, while Englehardt et al. [30] reported it on 1.6% of the top 1M sites. More recently, Iqbal et al. [39] proposed a dynamic detection approach based on JavaScript execution traces and reported fingerprinting signals on more than 10% of the top 100K websites. They also introduced API co-usage heuristics to distinguish fingerprinting from benign functionality.

Unlike stateful tracking, such as cookies or local storage, browser fingerprinting does not rely on storing identifiers on the client. Hence, it is more difficult to detect, and may persist despite browser-side restrictions on stateful tracking [34].

2.2 Client-side Security

JavaScript (&) Vulnerabilities: Web applications may be susceptible to a myriad of client-side vulnerabilities, such as Cross-site Scripting (XSS), Cross-site Request Forgery (CSRF), and Clickjacking, which may jeopardize the integrity of the application, leading to critical consequences. The majority of these client-side threats stem from insecure or untrustworthy third-party JavaScript, included by the web page itself, that executes in the trusted context of the visited web page and performs security-critical (in-)actions [47, 79]. For instance, a nefarious third-party JavaScript may steal sensitive cookies (cookie theft), overwrite them (session fixation), or perform other illegitimate stateful actions (CSRF, for instance) on behalf of the user. While the defense to most of these threats exists, applications often find it to be an obstacle to their functionality and difficult to migrate due to over-reliance on third-party JavaScript [74, 78]. To worsen the situation, inconsistent or insecure development practices often contribute to the threat surface in reality [11, 50, 82].

Security Headers: Modern browsers allow applications to

enforce a multitude of security policies and restrictions on the client side through HTTP response headers. That is, the application server may send HTTP response headers, including the desired security policy, along with the response content to the user. The browser parses these HTTP headers and imposes the desired restriction before rendering proceeds. There are 10+ security-related HTTP headers that allow fine-grained control of applications on the client side [63]. For instance, websites may send the `Content-Security-Policy` to block all unwanted third-party JavaScript through allowlisting, thus mitigating XSS-related vulnerabilities. Similarly, access to *cross-site* resources could be restricted through sending appropriate CORS headers. Websites may also selectively allow embedded content to request sensitive permissions (e.g., *camera*), by explicitly defining them in the `Permissions-Policy` header. Orthogonally, the `Strict-Transport-Security` header is meant to ensure secure and encrypted communication between the client and the server. Unfortunately, security misconfigurations are also quite common among these headers, and developers find it non-trivial to define desired security policies, as also reported by prior studies [17, 67].

2.3 S&P Practices across Geographies

A long tail of research has contributed to our understanding of the S&P development and deployment patterns among globally popular websites. However, we have a very limited understanding of applications that are only regionally popular or that cater services to local users. For instance, Böttger et al. [13] evaluated the effectiveness of filter lists across regional websites by visiting 9 countries using AWS services. They reported that the current state of filter lists does not fare against regional trackers. Orthogonally, Hu et al. [33] analyzed the behavior of trackers in multi-country settings to find that trackers are location sensitive and the top-ranked websites may overestimate their overall prevalence. Ogut et al. [62] investigated the behavior of cookie-consent banners among popular websites and reported that it is either not feasible or very complicated to reject cookies in most websites, across countries. Unfortunately, none of the prior studies investigate differences in the S&P postures of websites across regions in real-world settings. We aim to fill this research gap by comparing the behavior of global and regional websites under local conditions across 5 different countries in our study.

3 Research Methodology

In this section, we describe our research methodology and experimental setup to conduct a *representative* web S&P measurement that captures the true S&P-related *behavior* of websites across geographies. We investigate web applications that are popular and accessible from 5 different countries – *Algeria, Germany, India, the UAE, and the USA*, to showcase the feasibility and relevance of our approach. In the following, we

first detail on the website cataloging strategy that we adopted here and allowed for fair representation of global and local websites for corresponding country. Then, we describe our data collection infrastructure and also provide an overview of the S&P controls that we investigate across different website catalogs and countries. We provide a detailed rationale on the selection criteria of 5 countries for this study in Section 4.

3.1 Website Categorization & Cataloging

Contrary to prior studies, we consider three distinct catalogs of websites for our S&P measurements, to address our overarching research question: *how do Web applications differ in terms of their S&P representation and behavior across geographical regions?*. At a high level, the first catalog consists of globally popular websites, the second consists of country-coded versions of globally popular websites (e.g., [amazon.in](#) for India), and the third consists of websites that are primarily popular within the country. Note that while we only discuss our cataloging strategy for the 5 countries we selected here, the approach is extensible to any set of countries.

General Strategy. We construct these catalogs based on the top-level domains of websites. We rely on two different sources to construct lists of top-level domains (TLDs) associated with each country. First, we refer to the IANA Root Zone Database [35] from where we extract TLDs that are listed as country-code: `.dz` for Algeria, `.de` for Germany, `.in` for India, `.ae` for the United Arab Emirates, and `.us` for the United States. However, the IANA Root Zone Database does not cover recently-proposed geographic generic TLDs and therefore misses region- or city-specific TLDs (e.g., `.abudhabi` is listed as a generic TLD). To complement this, we refer to the ICANN gTLDs [36], which provides a list of applications for new geographic top-level domains. Please refer to Table 4 for the list of eTLDs for the countries that we considered.

Globally-popular websites (D1). This catalog consists of globally popular websites. We consider these sites from each vantage point to investigate whether users from different regions are subject to different levels of S&P attributes and risks when accessing the same set of websites.

To construct *D1*, we first fetch the top 10,000 global sites from CrUX [18]. Then, for websites that appear in the top 10,000 under multiple top-level domains (e.g., [amazon.com](#), [amazon.in](#), and [amazon.de](#) all appear in the global top 1,000), we exclude the versions with country-specific top-level domains corresponding to one of our vantage points we consider (e.g., `.de` for Germany, `.in` for India), as we analyze these separately in *D2*. In the example above, only [amazon.com](#) is included in *D1*. We apply this filtering because country-coded versions of global domains are explicitly tailored to users in the corresponding country and appear in the top 10,000 largely due to local Internet traffic. For instance, [amazon.in](#) ranks among the top 1,000 global websites due to its traffic volume within India, rather than global popularity [70].

Hence, due to its eTLD, [amazon.in](#) is only accounted for in the *D2* catalog for India. Thus, we ensure that there are no overlapping URLs between two (or more) countries in the *D2* catalog. After applying this filtering, 7,761 sites remain from the top 10,000. To complement this list, we randomly sample 2,239 additional websites from the top 50,000 global URLs, applying the same exclusion criteria.

Country-coded versions of global websites (*D2*). This catalog includes the country-coded versions of global websites. We include it in the study to investigate whether users accessing different country-coded versions (e.g., [amazon.de](#), [amazon.in](#)) of the same global website (e.g., [amazon.com](#)) encounter different S&P controls and risks.

To construct *D2*, we first generate a candidate set of country-coded versions of global websites. We fetch the top 1 million sites for each country, as well as the top 1 million global sites, from CrUX. For each country, we add a site to the candidate set if it satisfies three conditions: (1) it does not appear in *D1*, (2) it uses one of the country’s TLDs (e.g., [.in](#) for India), and (3) its second-level domain also appears among the global top URLs under a different TLD. The last condition ensures to identify country-coded versions of globally popular services. For example, we only consider [example.in](#) for India’s *D2* catalog if another version, such as [example.com](#) or [example.org](#), appears in the global top URLs.

Second, we validate these candidates to eliminate false matches and ensure that each site in the *D2* catalog is likely to be a country-coded version of a global service (e.g., [aadl.dz](#) and [aadl.com](#) share the same second-level domain but do not represent the same entity). For each candidate and its mapped global equivalent, we compare two signals: *favicons* [86] and *TLS certificates* [37]. Favicons provide a visual identifier of a website’s brand. When two domains use similar favicons, this suggests that they are operated as part of the same brand or service. We collect the favicon for each candidate and its corresponding global site, and compute their visual similarity using embedding-based image comparison. We also compare TLS certificate fields: Subject Alternative Names (SANs) and organization information. SANs list the domain names covered by a certificate; therefore, indicating whether the candidate and its mapped global equivalent are covered by the same certificate. Organization fields identify the entity to which a certificate was issued, and alludes towards shared ownership. We collect TLS certificates for all candidate *D2* sites and their mapped global equivalents, extract these fields, and compare them across each candidate. We retain a candidate in *D2* if it is validated by either favicon similarity or TLS certificate evidence (additional details in Appendix A).

Country-specific popular websites (*D3*). *D3* includes websites that cater services to the users of a corresponding country but may not rank among the popular sites globally. Studying these sites enables us to examine cross-country disparities and to gain insight into the S&P practices adopted by local developers, as they are more likely to be developed and

Table 1: Number of websites per catalog and per country

Catalog	Algeria	Germany	India	UAE	USA
D1	10,000				
D2	65	4,278	1,535	505	371
D3	2,981	10,000	10,000	9,749	9,122

maintained within the country. To construct *D3* for individual countries, we first fetch the country’s top 1 million websites from CrUX and exclude any site (hostname) already included in *D1*, and, by extension, *D2*. This ensures that websites in *D3* are predominantly popular within the country but not globally.

Next, we only retain websites that use one of the country’s top-level domains. Although some country-specific websites may use generic or non-country-specific TLDs, we chose to exclude them to ensure a clear distinction between globally popular sites and those intended primarily for local audiences. Finally, if the resulting set contains more than 10,000 websites, we select the top-ranked 10,000 domains per country to form the final *D3* dataset.

Table 1 presents the website counts across catalogs and countries. We apply a best-effort strategy to build large *D2* and *D3* catalogs across all countries; Nevertheless, the number of websites in *D2* and *D3* differs across countries. This variation reflects cross-country differences in the prevalence of country-associated TLDs among websites. For *D2*, a site must use one of the country-associated TLDs and be linked to a globally popular website under another TLD. Thus, smaller *D2* catalogs indicate that fewer global websites operate country-coded domains for individual countries. Whereas for *D3*, a site must appear in the country-level CrUX ranking and use one of the country-specific TLDs. Thus, smaller *D3* catalogs indicate that fewer sites use the corresponding country-specific TLDs. We resorted to a conservative strategy here to filter out any false positive matches, however, the approach is extensible under tolerable false positive thresholds.

3.2 Measurement Overview

Figure 1 describes the overall measurement methodology and the flow of our analytical pipeline. We begin by collecting and categorizing URLs based on worldwide popularity (*D1*), their indirect presence across countries (*D2*), and limited accessibility across individual regions (*D3*). We visit all the *D1* sites, along with the sites associated with individual countries in *D2* and *D3*, from respective countries, and store all the crawl artifacts and runtime behavior through our crawling pipeline. In the end, we analyze these crawl data to identify variances in their S&P behavior across URL catalogs and countries.

We perform the crawls using *PageGraph* [15], a web measurement tool integrated into the Brave browser [16]. *PageGraph* records and represents page execution as a directed graph in which nodes correspond to runtime entities and

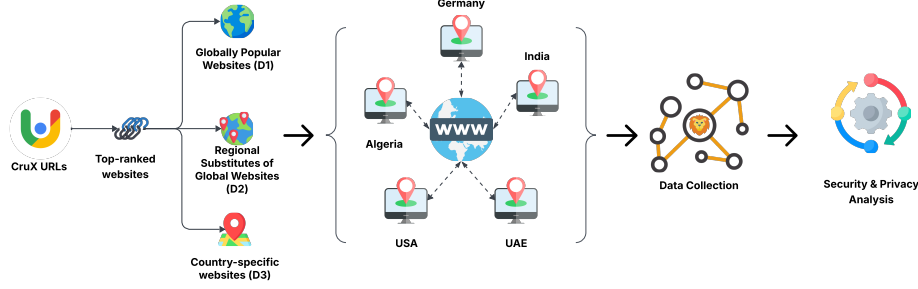


Figure 1: An overview of our URL sampling strategy, automated data collection, and analysis pipeline.

events, including scripts, DOM elements, JavaScript calls, and storage operations, and edges represent relationship events, such as element creation, script inclusion, and network requests. This representation enables precise attribution of recorded execution activity to specific scripts and execution contexts (e.g., identifying the script element responsible for creating a given HTML element). Since *PageGraph* does not record HTTP request headers [15], we complement our data collection by also recording HTTP Archive (HAR) files. HAR files include request and response headers and all other metadata associated with network requests.

We disable *Brave Shields* for all our crawls (i.e., all additional privacy protections turned off) to capture organic web activity [75]. Since Brave is Chromium-based, this setup provides a similar browser environment to that experienced by the majority of internet users [76]. For each site, we attempt to visit up to three pages to capture behavior on both landing pages and any further pages that it embeds [5]. We start with the landing page and iteratively select random same-site URLs, embedded within the landing page, until we visit three distinct pages. We stop crawls in case we do not find any embedded links on the landing page. In addition, to mitigate server randomness, we repeat each URL visit three consecutive times with distinct browser profiles and use the resulting observations from all three visits in our analysis. Moreover, inspired by previous works, we apply a best-effort strategy to accept cookie consent banners on each page [57]. We design our crawls to be stateless across websites: we create an individual browser profile for each website (including sub-pages) to avoid bias from state carried over across different websites. Lastly, we disable client-side caching via Puppeteer to ensure that all resources are fetched afresh for each page visit [64].

3.3 Privacy and Security Measures

We utilize the crawl artifacts to analyze websites S&P posture across regions. We consider common privacy-invasive behaviors (e.g., third-party tracking and browser fingerprinting), as well as client-side security configurations (e.g., security-related HTTP headers and script inclusion practices) for our study. For the latter, our goal is not to identify specific secu-

rity vulnerabilities among websites, but to examine broader deployment characteristics and associated variations across global and regional deployments. Thus, we focus on a set of security indicators that are not tied to specific vulnerabilities or website categories, but rather provide an overview of differences in the development and deployment of web applications across locations, following related works in this direction.

3.3.1 User Tracking

We identify tracking-related network requests using community-maintained EasyList filter lists. Specifically, we use EasyList [26], which targets advertising-related domains and resources, and EasyPrivacy [28], which focuses on non-advertising trackers such as analytics and user behavior profiling. In addition, we include several regional *Affiliated Filter Lists* [27], which capture tracking and advertising resources specific to particular linguistic or regional contexts. These include *IndianList* (for Indian-language websites), *Liste AR* (Arabic-language websites), and *EasyList Germany* (for German-language websites). We merge all filter lists into a single list and apply it across all crawls.

3.3.2 Browser Fingerprinting

We detect browser fingerprinting by analyzing JavaScript execution traces and identifying scripts that access browser APIs known to expose device- or browser-specific attributes. Our detection logic is rule-based and derived from prior work [39]. We consider several fingerprinting techniques, including canvas image fingerprinting, WebRTC-based fingerprinting, and AudioContext fingerprinting [1, 39]. We also match executed scripts against a list of known fingerprinting scripts by Iqbal et al. [39], using both script source URLs and script hashes.

3.3.3 Client-side Security Constraints

Web applications may restrict their behavior on the client side (i.e., on the user’s machine) by dictating security boundaries to the requesting browsers. That is, application servers themselves define these security policies and send them in

the form of HTTP response headers to the browser along with the requested content. Therefore, HTTP headers, such as *Content-Security-Policy* and *Strict-Transport-Security*, play a pivotal role in mitigating client-side vulnerabilities, such as cross-site scripting, clickjacking, and connection-downgrade attacks, to name a few. Thus, we consider the deployment of various security-related headers, along with their policy configurations, as key indicators to assess the security controls of individual websites across dimensions. Notably, the geolocations (the local governmental or censorship policies) we consider in our study have not been reported to influence the deployment of HTTP headers and their corresponding values in the past and are dictated by the applications themselves. Although websites send a multitude of headers, not all of them are relevant in the context of security (e.g., *Date*, *Etag*, *Server*, etc.). Hence, we only consider those headers which are prevalent and recognized to add security or privacy boundaries on the client-side [59, 63], and filter out the rest. This way, we select 22 headers we observe in our data, as listed in Table 5. We, then, analyze the distribution of individual headers across websites and evaluate the effectiveness of associated policies. Through this analysis, we compare the adoption and behavior of these headers across regions, in line with previously studies [55, 68, 80].

3.3.4 Script Inclusion & Execution Patterns

While HTTP headers indicate the security configuration desired by websites, requests may also include resources from untrustworthy or vulnerable third-party sources, which may compromise the integrity on the client side. For instance, a CSP header with a trusted `script-src` origin may allow a vulnerable or (later-)compromised JavaScript file that exfiltrates sensitive data or perform critical state-changing actions on users' behalf. This behavior is particularly dangerous and is widely reported in prior studies [66, 87]. Thus, we also consider the JavaScript sources, their runtime behavior and associated security policies (i.e., CSP directives), included by websites when compared to related protective mechanisms as a secondary signal to understand their security posture and related development practices. In particular, we capture the URLs of the scripts directly embedded by the visited URL or indirectly through any same- or cross-origin *iframes*. Additionally, we also monitor all dynamic script injections at runtime through *PageGraph*. We then aggregate these script sources based on the visited URL (top domain) and analyze their inclusions and runtime behavior across geographies. Subsequently, we also monitor JavaScript API invocations at runtime along with their source (i.e., first- or third-party scripts).

4 Country Selection & Configuration

A crucial element of our study is to obtain website behavior as close to the users' reality as possible. However, this constraint

also affects our country-selection strategy and scale. In the following, we first explain our rationale on country selection and associated limitations. Next, we detail on a small-scale measurement experiment which confirms the influence of VPN and cloud services on websites' behavior.

4.1 Choice of Vantage Points

We selected the 5 countries – *Algeria, Germany, India, the UAE, and the USA*, for our study due to two critical reasons.

First, they include countries that have been heavily studied in prior work (Germany [13, 62] and the USA [1, 30]) as well as understudied countries, albeit, with significant Web traffic (Algeria, India, and the UAE), with a total share of 30% global web traffic [4, 21, 77]. Additionally, due to access, legal, and censorship concerns, we decided against running our study in China and Russia [46, 88, 90].

Secondly, since our study design necessitates conducting measurements under "real-world conditions", we relied on using machines that were physically located in these countries with and local network conditions to maintain a realistic measurement setup, and hence, limited our study to 5 countries. This is because various research evidence suggests that VPN (or similar) traffic is often fingerprinted by websites, leading to *altered behavior* when accessed through these services [13, 41]. Using local machines reduces this risk and ensures that websites behave as they would for real users. Naturally, we believe this to be an acceptable trade-off in terms of overall representation and contribution through our research.

4.2 Influence of VPN & Cloud Traffic

Motivated by our underlying research question, we investigate and report our findings on the apparent *behavioral* differences between VPN and non-VPN traffic. That is, we conduct a small-scale experiment to identify and evaluate the differences among websites' content and behavior when accessed over VPN, compared to local network conditions.

We randomly select 1,500 globally popular websites (D1) and crawl each site both from physical machines in each country, and through VPNs, following the methodology presented in Section 3.2. We use a single vantage point to simulate connections from Algeria, Germany, India, the UAE, and the USA using VPNs. Notably, to ensure that the VPN tunnel's exit node is physically located within the target country rather than mimicking a virtual address assigned by the service provider, we use different service providers for individual countries as required. This way, our setup is not prone to conditional redirection based on the *location* header or partial view due to VPN-driven routing. Specifically, we use NordVPN [61] for Germany, the USA, and the UAE; VyprVPN [85] for Algeria, and Windscribe [89] for India. Additionally, we disable all optional VPN protections (e.g., tracking or ad-blocking). While

certain VPN or cloud service providers may allow further network configurations, we resort to the default configuration for a uniform setup environment. We compare the traffic recorded from physical vantage points against VPN traffic across individual countries. For each crawled URL, we compare the: (1) average number of network requests, (2) average number of third-party script components, (3) JavaScript API invocations, and (4) security-related response headers sent by websites.

Overall, 713/1,157 (61%) URLs accessible across all five countries showed significant differences when accessed through VPN in at least one of these countries. We observe divergent effects of VPNs on the total number of network requests across countries, depending on the median number of requests per site. Websites in the USA showed no effect over VPN (214/217). VPN traffic in Germany also showed negligible differences (383/372), indicating that some sites loaded more content over VPN than over non-VPN traffic. This trend was more pronounced in India, where sites loaded more content over VPN (120/375), potentially bypassing any blocking or filtering mechanisms in place. Whereas, VPN traffic in the UAE strongly suggests that less content was loaded (429/398) than through normal access.

Next, we investigate third-party scripts loaded by websites across VPN and non-VPN accesses and observe similar trends in overall network requests. 72% websites in India load more third-party resources than the baseline access. For instance, [tiktok.com](https://www.tiktok.com) and [preferencenail.com](https://www.preference.nail.com) were accessible only via VPN on 42 sites. On the contrary, VPN access in Germany blocked ad-related scripts on over 81% sites, such as [criteo.net](https://www.criteo.net) and [adtrafficquality.google](https://www.adtrafficquality.google). Figure 2 shows the effect (gain/loss) using VPN on popular third-party scripts. 69% websites in UAE showed less ad-related scripts over VPN, while some websites now included bot-detection scripts (e.g., [cloudflareinsights.com](https://www.cloudflareinsights.com)). As before, accesses in the USA were least affected, where popular ad-related scripts were suppressed while others were loaded (e.g., [tapioni.com](https://www.tapioni.com)). Websites also include third-party iframes to execute scripts. However, we observed similar trends with VPN as with scripts, detailed in Figure 4. Moving forward, we measure JavaScript API invocations via these third-party scripts. As before, the API invocations in India and Algeria show higher tracking (+4.5%) and fingerprinting (+11.6%) over VPN, whereas we observe fewer invocations in Germany (-6.9%). Interestingly, while fewer scripts were loaded over VPN in the UAE, we record higher JavaScript invocations (+6.5%). Figure 3 summarizes the distribution of the JS API over VPN across different countries.

Lastly, we do not observe significant deviation for most security-related headers between VPN and local network conditions, as listed in Table 6. A lower number of cookies were transmitted via the `Set-Cookie` header when using VPNs, except for a negligible rise in India. We also observe a weak, unidirectional trend of fewer security-related response headers over VPN in the UAE, USA, and Germany compared to

India. We believe the VPN service we used in the former three countries for measurement could also contribute to the observed deviations here. Given that this analysis is limited to landing-page visits of 1,500 globally popular websites, these results suggest that VPN-based measurements may be sufficient for studying header-related deployments in isolation.

We do not aim to extensively analyze differences between VPN-based and non-VPN measurements, the reasons behind these discrepancies, or the performance of specific VPN services. Nevertheless, these results highlight that relying on VPN-based measurements provides a limited perspective compared to users' experience. This motivated our decision to rely on physical machines in each location for our study.

5 Evaluation & Results

In this section, we report on our findings from large scale S&P measurement on websites across catalogs and geographic vantage points. In particular, we compare security-related practices (e.g., deployment of security-related HTTP headers and inclusion of third-party scripts) and the prevalence of privacy-invasive behaviors (e.g., browser fingerprinting and third-party tracking), as discussed in Section 3.3. We systematically analyze these behaviors using HAR files and Page-Graph's graph-structured representation of execution logs, which captures script execution and inclusion chains.

Overall, across all catalogs, we crawled 13,046 sites from Algeria, 24,278 from Germany, 21,535 from India, 20,254 from the UAE, and 19,493 from the USA. The overall success rate is 86%. By country, the success rate is 81% in Algeria, 84% in Germany, 88% in India, 89% in the UAE, and 84% in the USA. The remaining failures were primarily due to HTTP 404 responses and timeouts.

5.1 State of Security-related HTTP Headers

We first discuss the variance in the deployment of security-related headers on popular global and regional websites.

5.1.1 Overall Trends in Deployment

We consider the presence of security-related HTTP headers (or lack thereof) and their configuration as an indicator of the security posture of individual websites. Hence, we identify the most common occurring set of security-related response headers and report their adoption across different countries and site catalogs. Importantly, we consider a header deployed by a website only if it consistently appears across all 3 visits to the corresponding URL. Table 7 details the overall distribution of the 22 most frequently occurring headers across three different catalogs of websites, respectively.

Notably, we find several patterns and anti-patterns spanning across the three URL catalogs and countries. To begin with, the overall trend of deployment increases unidirectionally

from *D1* to *D3* for more popular headers such as *CSP* (except Algeria and the UAE) and *HSTS*, whereas this is inverse for recent additions (e.g., *COOP* and *Report-To*). Then, the fraction of deployments of individual headers also appears to differ selectively across global websites (*D1*), where we observe a general trend of slightly higher occurrences in *UAE*, followed by *Germany* and *Algeria* (interchangeable for different headers). On the contrary, the same set of websites sent fewer headers (or sent them inconsistently), when accessed from India, and the least in the USA – approximately 45% fewer than in the UAE for each header, directing that the security controls are not uniform across the same set of websites and are heavily dependent on users’ access points.

The differences are more pronounced among *D2* and *D3*, where we see inconsistent trends across individual headers and countries. In general, *X-CTO* (34.5%), *HSTS* (36.9%), and the (now-deprecated) *X-FO* (34.1%) headers remain among the prevalent ones across all dimensions. In contrast, recently introduced ones, such as *COOP* and *CORP*, are widely visible across *D1*, while *Permissions-Policy* and *CORS* headers are more prominent in *D2* and *D3*. India-specific sites are an interesting outlier: of the sites that sent the *ACA-O* header, only 53.9% of them included wildcards (compared to other *D3* sites with 68–77%), while a large fraction of them specifically allow-listed the requesting origin.

Overall, the distribution of headers shows that focusing solely on global URLs (*D1*) may provide early indicators of usage, but it does not accurately reflect the state of security header adoption in practice.

5.1.2 Effective Policy Configurations

While websites may deploy security-related headers, a weak, misconfigured, or erroneous policy may render them useless. As an example, an *X-FO* header set to (deprecated) `ALLOWALL` or a *HSTS* header with `max-age` set to 0 is effectively ignored by the browser. Thus, we assess the effectiveness of a subset of these headers by analyzing the values and directives they include. We leverage the strategy proposed by Roth et al. [68] for four popularly deployed headers – *CSP*, *X-FO*, *HSTS* and *Set-Cookie*. While Roth et al. originally used this strategy to study inconsistencies in header deployment and enforcement across variable client-side configurations (*only*) over VPNs, our goal is orthogonal: we use the formal constraints modeled by their strategy to evaluate header configurations across global and regional websites and across countries.

To measure effective policy enforcement, we first normalize individual header values before applying the quality checks. When a single response contains multiple *HSTS* headers, we consider only the last occurrence; for *CSP*, we concatenate all *CSP* header values. Tables 8 to 11 summarize the effective quality assessment for the four individual headers. For *CSP*, *X-FO*, and *cookie policies*, we only consider cases where the header is consistently served with identical values across all

three visits. For example, we exclude cases where a website sends the *X-FO* header with different values across visits, and therefore report conservative observations. For *HSTS*, a single secure deployment suffices.

Overall, *CSP* deployment and policies vary substantially even when the same sites are accessed from different vantage points. We find 7.51% of *D1* sites *consistently* serve different *CSP* policies across locations. Among these, 4.2% differ in the set of directives they included, while 3.88% also deviate in the set of allow-listed sources. We also find that 3.72% of sites differ in terms of enforcement (*CSP-RO* vs *CSP*). One notable pattern we saw unique to USA-specific catalogs was that they allow-listed native-app-schemes (`'safari-extension:'`, `'chrome-extension:'`, `zoommtg`, and `rtmp`;) on over 309 zoom sub-domains (`*.zoom.us`), unreported by prior works.

In addition to the four headers above, we also investigate differential policies in the *Permission-Policy* header, which controls access to sensitive client-side features such as geolocation and sensor APIs. In general, using Cloudflare Radar site categorization [19], we find that this header is majorly shipped by adult, gambling, piracy and streaming sites across all catalogs and countries (e.g., `ara.xhexperience.xyz` in *D1*, `memesapk.co.in` in *D2*, and `amankanliga365.us` in *D3*). Moreover, we find two notable cross-country cases worth mentioning here: (1) India-specific (*D3*) sites aggressively restricts (60.8–79.4%) camera, geolocation and microphone API permissions through explicit policy. This contrasts with other countries, especially the UAE (43–53%) and Germany (38.2–42.3%). (2) This trend was even more vivid when only considering government sites: most UAE government sites (`*.gov.ae`) allowed wildcard access to sensitive APIs, whereas Indian government sites (`*.gov.in`) explicitly shipped denying policies to users.

Overall, these results show that header deployment varies across catalogs and vantage points, and deployed headers can differ substantially in their effective configurations. In particular, country-coded (*D2*) and locally popular (*D3*) websites expose policy patterns that are not captured by globally popular websites alone, while even the same *D1* websites may serve different policies from different vantage points. These findings suggest that robust S&P measurements should evaluate both the presence and the effective quality of security headers, while accounting for website catalog composition and geographic vantage point.

5.2 Third-party Script Inclusions

Websites often include JavaScript from numerous third-party sources to integrate services and features on the client side. However, these third-party JavaScript libraries could become malicious, untrustworthy, or even vulnerable, exposing the application to threats. Thus, we consider the prevalence of these JavaScript sources an additional signal of an application’s security posture. We analyze the distribution of first-party

vs. third-party JavaScript across websites for each vantage point and website catalog. We define *first-party* (1P) scripts as scripts loaded from both *same-origin* and *cross-origin-but-same-site* sources, while *cross-site* sources are naturally labeled as *third-parties* (3P). We present the top 20 third-party origins and their prevalence across websites in Table 12. Additionally, we investigate the JavaScript inclusion chain and the script execution context to identify third-party *iframes* that include JavaScript. We only consider websites and scripts that showed consistent behavior across three different visits.

D1: Global Websites and Trends. Over 92.7% of *D1* sites include at least one third-party script, with a median of 75 3P-scripts from 4 unique domains. This held nearly steady at 92.4% for *D2* sites, however, at a median of 121 scripts from 4 domains, but dropped to 77.0% for *D3* sites, with a lower median of 60 scripts from 3 domains. This already indicates that the prevalence of third-party scripts is non-uniform across global and local websites. This trend is also visible at the level of individual 3P-script domains. For instance, [googlesyndication.com](#), one of the top occurring 3P-script domain appears on 20.54% of *D1* sites, compared with only 7.23% of *D2* and 5.37% of *D3* sites. Similarly, [criteo.net](#) appears on 8.78% of *D1* sites, versus 1.89% in (*D2*) and 1.07% in (*D3*), strongly indicating that their presence and behavior on the web may be overrepresented when only focused on a global set of websites. We also observe vantage-point differences within *D1*. Over 8.6% of global websites load scripts from 10 or more third-party domains based on the point of access. For example, the scripts loaded from the *AliExpress* sub-domains ([*.aliexpress.com](#) appeared in Germany and the UAE more frequently (75%–95% share) and with high recursive inclusion depth (5) than what we observe in the USA (33% share) with only direct inclusions.

The UAE Anomaly: We expand our analysis of 3P-script request volume per website by taking the vantage point into consideration, as detailed in Table 13. Here, we see clear distinction in trends based on the point of access, as well as the website catalogs. The UAE is an outlier in dual directions: the website behavior flips between the high outlier and the low outlier based on the URL category. For globally popular websites (*D1*), visits from the UAE trigger higher numbers of 3P-scripts (2.52x compared to Germany) from specific set ad-exchange sources. For example, on [9db.jp](#), a single 3P-script domain, [doubleclick.net](#), recursively embedded six more scripts downstream through `document.write`, which we did not observe in other countries. Whereas, this overall behavior flips the other way for websites in *D2* and *D3*. Websites in Germany loaded 2x ad-related scripts than the UAE counterparts. We speculate and attribute this behavior to exceptionally high Internet penetration rate and unique local ad ecosystem in the region [43].

The above findings distorts our existing understanding of third-party script inclusion patterns, the associated behavior (e.g., ads) and threats (e.g., supply-chain vulnerabilities), and

means to measure them at scale, as prior work in this direction sheds light on [78, 79, 80].

D2/D3: Regional Influences on Websites. We showed that globally popular websites load different third-party scripts and at different rates, based on the access point. Regional and country-coded websites show similar trends where they even go further to include scripts from regional services over global players. We saw this explicit trend among websites that loaded AI-assistance related scripts. That is, while only 1.60%–2.11% *D1* websites loaded them, this starkly increased for *D2* (2.45%–7.8%). Interestingly, the UAE sites in *D2* and *D3* were the outliers here as well which included these script the most, compared to other countries (7.8% and 2.83%). [tabby.ai](#), for example, is an AI-powered buy-now-pay-later checkout assistant that appeared only on 12 distinct UAE-specific storefronts (e.g., [michaelkors.ae](#) and [underarmour.ae](#),). Similarly, [dynamicield.com](#), an AI-personalization tool, showed up (only) on multiple UAE-specific retail sites (e.g., Swarovski and Lacoste). While Germany sites also showed elevated inclusion of AI scripts in *D2* (3.71%), the trend is spread over multiple vendors.

Next, we performed a deeper analysis at the JavaScript activity of the embedded 3P-scripts at runtime and report interesting deviations. Here, we categorized closely-related JS APIs, reported by prior works to be security-critical [6, 7, 8, 38, 72, 73]. We saw clear downhill activity in eval and service-worker related API calls ([onesignal.com](#) being one prominent case) among *D2* sites when compared to *D1* sites, across all countries. Contrastingly, the fingerprinting and network-related JS activity (in [clarity.ms](#) and [doubleclick.net](#) for example) was more prevalent. Irrespective of the observed effective fingerprinting rate, as discussed in Section 5.4, the associated API calls in the UAE was almost 2.2 times more than other countries among *D2* sites. Since the findings strongly suggest that third-party scripts and their runtime activity are heavily customized to the accessors’ location, merely assessing their true behavior and associated threats and vulnerabilities from a single vantage point may be an underestimation.

Parallel to third-party scripts, we also analyzed the prevalence of 3P-*iframes* and saw mirroring trends. That is, 82.72% *D1*, 86.86% *D2*, and 80.18% *D3* sites loaded at least 3P-*iframes*. However, the unique 3P-domains counts were 4x higher in *D1* across countries, with highest concentration in the UAE. For example, [theguardian.com](#) included 278 3P-*iframes* in the UAE, while none when accessed from Germany. Furthermore, the *iframes* among *D2/D3* websites heavily skew towards CMP and social-media widgets, as opposed to global websites that concentrates largely around ad-exchange infrastructure. Contrastingly, the USA-local websites included *iframes* from least diversity of third-parties. For example, [consentmanager.net](#), [usercentrics.eu](#) and [paypal.com](#) showed up in 100+ Germany-specific sites, but none of them appeared in their USA-counterpart.

Overall, our findings corroborate the fact that regional sites

Table 2: Overview of third-party tracking across website catalog and vantage point. (a) prevalence (% of sites with at least one tracker); (b) average number of distinct trackers per site; (c) total distinct tracking domains across all sites.

	Algeria	Germany	India	UAE	USA
<i>(a) Tracking prevalence (% of sites)</i>					
<i>D1</i>	71.5	70.3	68.4	71.2	70.6
<i>D2</i>	73.3	74.2	71.5	76.0	67.1
<i>D3</i>	35.0	50.8	36.2	65.1	55.4
<i>(b) Avg. distinct trackers per site</i>					
<i>D1</i>	12.6	13.8	9.2	16.0	11.5
<i>D2</i>	5.2	7.9	6.0	9.9	8.0
<i>D3</i>	2.7	8.5	4.8	5.3	4.3
<i>(c) Total distinct tracking domains</i>					
<i>D1</i>	1,985	2,008	1,747	1,993	1,926
<i>D2</i>	55	980	482	314	236
<i>D3</i>	137	1,025	396	465	639

may serve as a more potent starting point to measure the web supply-chain and region-specific deployments. More generally, polymorphic behavior of sites and the resources that they embed also necessitates comprehensive analysis by considering these dimensions for holistic assessment.

5.3 Third-party tracking

We next analyze user tracking across different site catalogs and countries. We detect tracking as described in Section 3, and focus on third-party trackers. A third-party tracker, as opposed to a first-party tracker, is loaded from a site different from the website’s own site (eTLD+1). We focus on (i) how tracking prevalence differs across catalogs, (ii) how tracking varies geographically for the same *D1* websites, (iii) how country-coded *D2* sites differ from their global equivalents, and (iv) whether the set of trackers differ across locations.

Prevalence of third-party tracking. Table 2 summarizes tracking activity across catalogs and countries. We consider a site to perform tracking if any third-party tracker appears in any of its crawls, and consider a tracker present on that site if it appears in at least one crawl. The table shows that the *D1* and *D2* catalogs exhibit consistently higher tracking prevalence than *D3*. In *D1*, between 68.4% and 71.5% of sites load at least one third-party tracker across all the five countries. In *D2*, prevalence ranges from 67.1% to 76.0%. In contrast, *D3* shows a lower and more variable prevalence of tracking between vantage points. Third-party tracking appears on only 35.0% of Algerian and 36.2% of Indian *D3* sites, but on 55.4% of USA and 65.1% of UAE *D3* sites.

The apparent variations we report suggest that globally popular websites and their country-coded versions rely more heavily on third-party advertising and analytics infrastructure. Whereas, tracking on locally popular websites is less prevalent and varies more across vantage points, likely reflecting differences in local tracking and advertising practices.

D1: Geographic variation in tracking behavior. For *D1* catalog, we visit the same set of URLs from all five countries. This enables a direct comparison of tracking behavior across geographies while keeping the visited website constant. Table 2 shows that tracking prevalence in *D1* varies by only 3.1 percentage points across vantage points. This suggests that geographic location has limited effect on whether a globally popular site performs third-party tracking at all. However, we find that vantage point affects both the number and the set of trackers loaded when visiting *D1* sites. Table 2 shows that the number of distinct third-party tracking domains on *D1* sites varies by location. The UAE exhibits the highest average number of distinct trackers per site in *D1* (16.0), followed by Germany (13.8), Algeria (12.6), the USA (11.5), and India (9.2). Notably, India also showed the lowest total number of distinct tracking domains in *D1*.

We also compare the tracker set observed on the same site when loaded across countries. For each pair of countries, we consider every *D1* site and compare the distinct third-party tracker domains observed from the two locations. Depending on the pair of countries, only 37.4% to 48.3% of sites load exactly the same set of trackers from both locations. In contrast, 14.6% to 17.2% load completely disjoint sets, indicating that none of the trackers observed from one location appear when the same site is visited from the other. We identify recurring cases where specific trackers appear on *D1* sites only when visited from specific vantage point but not from others. For example, Spadsync trackers (spadsync.com) appear on 191 *D1* sites when loaded from the UAE, but not when the same sites are loaded from Algeria, Germany, India, or the USA.

In general, it is challenging to narrow down the exact cause as to why a tracker is visible from one location but not another, since various hidden (overlapping) factors may influence the application behavior: data-protection regulations, intended logical segmentation by the first-party itself, or differential execution of embedding third-party scripts. We manually inspect selected cases and identify one such example where the location-specific logic is visible in client-side JavaScript. When visiting <https://www.finanzen.net/ratgeber/garantie-zertifikate/> from Germany, the page loads the third-party tracking script <https://script.ioam.de/sensor/latest/manager/base/es6/bundle.js>. The same script is not loaded when visiting the page from Algeria, India, the UAE, or the USA. Manual inspection showed that this tracker is included by a cross-origin script loaded from <https://data-fdbbf15b66.finanzen.net/iom/latest/manager/base/es6/bundle.js>. This script implements country-specific logic: it restricts the tracker to Germany (de) and Austria (at), and selects a country-specific endpoint for each case (Listing 1).

D2: country-coded sites versus global equivalents. For each country, we compare the *D2* catalog with the corresponding set of global equivalents from *D1*, loaded from the same vantage point. For example, when analyzing Germany, we com-

```

1 var Ha = oa(["de", "at"]);
2 cdn: {
3   host: ({ country }) =>
4     country === "at" ? "script-at.iocnt.net" : "script.ioam.de"
5 }

```

Listing 1: Country-based tracker inclusion logic. The script includes different trackers from German and Austrian users.

pare country-coded sites such as amazon.de with their global equivalents, such as amazon.com, when crawled from the Germany. This comparison allows us to test whether users visiting country-coded versions of global services are exposed to different third-party tracking behavior than users visiting the corresponding global versions from the same location.

We find that tracking prevalence is high for both country-coded sites and their global equivalents, with no considerable difference between the two sets across countries. The prevalence values for *D2* sites and their global equivalents are, respectively: Algeria, 73.3% vs. 71.5%; Germany, 74.2% vs. 70.3%; India, 71.5% vs. 68.4%; the UAE, 76.0% vs. 71.2%; and the USA, 67.1% vs. 70.6%.

However, the trackers loaded by country-coded sites and their global equivalents often differ. For each country-coded site, we compare the set of distinct third-party tracker domains observed on that site with the set observed on its global equivalent. In 48.3% to 67.4% of cases, at least one tracker appears only on the country-coded site. Contrastingly, in 54.0% to 71.3% of pairs, at least one tracker appears only on the global equivalent. For instance, across Germany, India, and the UAE, we record the tracking script bat.bing.com/bat.js only on the country-coded version of 21 websites (e.g., www.parcelmonkey.de), and not on their corresponding global equivalents (e.g., www.parcelmonkey.com).

Top tracking domains across vantage points. Table 14 presents the top third-party tracking domains across *D1*, *D2*, and *D3* catalogs. The table shows that the same set of Google advertising and analytics services dominates the tracking infrastructure across all vantage points: googletagmanager.com appears on 37.2%–53.0% sites, google-analytics.com on 27.5%–37.8%, doubleclick.net on 21.8%–37.0%, and google.com on 21.0%–35.5%.

However, the total number of distinct tracking domains observed on all websites varies across vantage points. Table 2 shows that Germany has the largest tracker ecosystem, with 2,008 distinct tracking domains in *D1*, 980 in *D2*, and 1,025 in *D3*. Algeria has the smallest, with 1,985 in *D1*, 55 in *D2*, and 137 in *D3*. Moreover, we identify trackers that are unique to individual vantage points; the top 5 per country are presented in Table 15. The most prevalent are localized Google domains: google.dz in Algeria (17.5%), google.de in Germany (16.8%), google.co.in in India (13.7%), and google.ae in the UAE (20.8%). Other country-exclusive trackers are much

less prevalent, typically appearing on around 1% of sites in their respective countries, such as sparkasse.de in Germany (1.01%) and tpmn.co.kr in the USA (0.82%).

5.4 Browser Fingerprinting

We finally report on the prevalence and geographic variation among *browser fingerprinting* behavior, a tracking technique that leverages browser APIs to derive high-entropy user identifiers without relying on client-side storage.

Prevalence of browser fingerprinting. Table 3 reports the percentage of websites exhibiting fingerprinting behavior across website catalogs and vantage points. We consider a site to perform fingerprinting if we observe the associated activity in at least one visit to the site. Similar to third-party tracking, fingerprinting is most prevalent on globally popular websites. In the *D1* catalog, 15.8%–20.4% of sites perform fingerprinting; the *D2* catalog shows lower prevalence, from 6.7%–17.0%; and the *D3* catalog exhibits the lowest prevalence, from 0.5%–6.2%. The geographic pattern is not uniform across catalogs. In *D1*, Germany and Algeria have the highest fingerprinting prevalence (20.4% and 20.2%), while India has the lowest (15.8%). In contrast, the UAE has the highest prevalence in both *D2* and *D3* (17.0% and 6.2%), while Algeria has the lowest (6.7% and 0.5%).

Fingerprinting techniques. Table 3 breaks down fingerprinting by technique. Across catalogs and vantage points, fingerprinting is primarily driven by canvas image and audio APIs. The prevalence of these two techniques mirrors the overall prevalence of fingerprinting: both are highest in *D1*, lower in *D2*, and lowest in *D3*. In *D1*, canvas image is the most prevalent technique, appearing on 11.5%–15.1% of sites, followed by audio fingerprinting at 8.1%–12.4%. This trend holds for *D2*, where canvas image appears on 5.0%–13.1% of sites and audio on 3.6%–8.6%. It also holds in *D3*, but at lower prevalence: canvas image appears on 0.5%–5.4% of sites and audio on 0.4%–3.3%.

WebRTC fingerprinting shows a different pattern. It is less prevalent overall, recorded on 0.0% to 5.1% of sites across catalogs, but it is more geographically uneven. In general, WebRTC is more prevalent in Algeria, the UAE, and India, and much less prevalent in Germany and the USA.

For instance, in *D1*, we observe WebRTC fingerprinting on 5.1% of sites from Algeria, 4.9% from the UAE, and 2.9% from India, compared with only 0.6% from Germany and 0.4% from the USA. We identify 188 *D1* sites where we record WebRTC fingerprinting from Algeria, India, or the UAE, but not from Germany nor the USA, including author.today, habr.com, and ria.ru. In all these cases, the WebRTC APIs are invoked by third-party *Yandex-related scripts*, such as <https://mc.yandex.ru/metrika/tag.js> on 143 sites, <https://mc.yandex.ru/metrika/watch.js> on 16 sites, and <https://cdn.jsdelivr.net/npm/yandex-metrica-watch/tag.js> on 9 sites. We also find that the

Table 3: Fingerprinting prevalence (% sites) per catalog and country (for all websites with detected behavior in any crawl).

	Algeria	Germany	India	UAE	USA
<i>(a) Overall prevalence by website catalog</i>					
D1 catalog	20.2	20.4	15.8	19.6	18.0
D2 catalog	6.7	9.4	8.6	17.0	14.8
D3 catalog	0.5	5.0	1.8	6.2	5.8
<i>(b) Per-technique prevalence by website catalog</i>					
<i>Audio</i>					
D1 catalog	11.6	12.4	8.1	10.0	10.2
D2 catalog	6.7	4.7	3.6	6.5	8.6
D3 catalog	0.4	3.3	1.1	2.3	2.6
<i>Canvas image</i>					
D1 catalog	14.3	15.1	11.5	14.8	13.0
D2 catalog	5.0	8.0	6.7	13.1	12.4
D3 catalog	0.5	3.7	1.6	3.9	5.4
<i>WebRTC</i>					
D1 catalog	5.1	0.6	2.9	4.9	0.4
D2 catalog	0.0	0.9	1.3	4.6	1.4
D3 catalog	0.0	0.4	0.0	1.8	0.1

same scripts are loaded when visiting these websites from Germany or the USA, but they do not execute WebRTC fingerprinting. This suggests that these scripts include conditional logic, potentially based on geographic location, that determines whether WebRTC fingerprinting is executed.

We manually inspect <https://mc.yandex.ru/metrika/tag.js>, as recorded during our crawls, to identify such logic. Although the script is obfuscated, we find that it implements WebRTC fingerprinting by creating an `RTCPeerConnection`. Notably, this behavior is triggered by a regional condition. The WebRTC routine is executed only when the internal `isEU` state is equal to 0 (Listing 2). While this condition suggests that the script distinguishes EU from non-EU visitors, we also observe that the routine does not execute from the USA.

Overall, this explains why the same Yandex script can be loaded from Germany without executing WebRTC fingerprinting, while executing it from Algeria, India, or the UAE. One possible explanation is that this behavior reflects region-specific privacy or data-protection configurations, such as those related to Yandex’s GDPR handling [91].

Statistical tests and validation. We use statistical analyses to validate the geographic differences observed in third-party tracking and fingerprinting. Specifically, we examine whether differences in (1) tracking prevalence, (2) fingerprinting prevalence, and (3) the number of third-party trackers across locations are statistically significant.

In *D1*, we visit the same set of websites from all five vantage points. We apply Pearson’s chi-squared tests to determine whether location has impact on tracking and fingerprinting prevalence, and the Friedman test to compare third-party tracker counts across locations. We find no significant difference in tracking prevalence ($\chi^2, p = .224$), whereas fingerprinting prevalence differs significantly ($\chi^2, p < .001$). The Friedman test shows that tracker counts also differ significantly across vantage points ($Q(4) = 301.43, p < .001$).

```

1 function collectWebRTC(window, telemetry) {
2   const PeerConnection = window.RTCPeerConnection
3   if (window.navigator.onLine && PeerConnection) {
4     const pc = new PeerConnection({ iceServers: [] });
5     // WebRTC fingerprinting
6   }
7 }
8 if (state.get("isEU") === 0) { collectWebRTC(window, telemetry); }
  ↪ // Conditional call

```

Listing 2: Simplified reconstruction of the regional condition controlling WebRTC fingerprinting in Yandex Metrika.

In *D3*, each vantage point contains a distinct set of websites. Hence, we apply Pearson’s chi-squared tests to determine whether tracking and fingerprinting prevalence differ across locations and the Kruskal–Wallis (KW) test to determine whether the distributions of third-party tracker counts differ across the independent samples. We find significant differences across locations in tracking prevalence ($\chi^2, p < .001$), fingerprinting prevalence ($\chi^2, p < .001$), and tracker-count distributions (KW, $p < .001$). The corresponding 95% confidence intervals are presented in Table 16. These confidence intervals support the prevalence differences we reported earlier, and indicate that the prevalence of tracking and fingerprinting, as well as the distribution of third-party tracker counts, varies significantly among the country-specific website samples.

6 Discussion & Conclusion

We now discuss the overall limitations, and broader implications of our research in the following.

6.1 Limitations and Threats to Validity

Bot detection. Despite using realistic network environments and physical machines, sites may still detect automated crawlers and alter their behavior. Such cases may lead to under-reporting of the prevalent behaviors. While we resorted to best-effort strategy to filter any bot-induced noise during crawls and analysis, certain mechanisms (e.g., Cloudflare Challenges [20]) may have evaded our scrutiny.

Instrumentation effects. We used *PageGraph* to monitor the runtime characteristics of websites here. Naturally, our framework subsumes the inherent limitations of *PageGraph*. Specifically, we noticed page crashes and load timeouts on content-heavy pages, induced by *PageGraph* instrumentation. To mitigate this side effect, we visited each URL up to 5 times until successfully visiting them 3 times. In addition, we used data from 3 consistently successful crawls in our analysis to account for server-side randomness.

Consent-banner interaction. Interacting with consent banners at scale is non-trivial. We used AutoConsent [24]

to interact with consent-management interfaces through site-specific rules and heuristics, following related web measurement studies [53, 57, 93]. However, this interaction remains a best-effort strategy, as AutoConsent may miss or incorrectly handle consent banners on some sites.

Incomplete detection and cloaking. Finally, we note that filter lists, while an invaluable source for annotating tracking domains and continuously updated, are also often incomplete or inadequate for identifying all tracking domains in the wild, including regional trackers. Thus, we believe that our overall findings only present a conservative lower bound of differences among websites across dimensions.

Our comprehensive measurement design allowed us to identify trends that correlated with geographical variations. However, geography, as a whole, subsumes several other confounds (e.g., regional data regulations, development norms, digital market) that may be responsible for the observed trends and variations. While we discussed interesting cases and potential influencing factors for reported trends, unfortunately, identifying exact root cause(s) behind these variations would require us to test each of the potential sub-factors in isolation to arrive on concrete causal factors for individual trends. Notably, this is an inherent issue with measurement studies in general and separate warrants qualitative investigation. Hence, we refrained from presenting uneducated causal relations within the scope of our study and leave it for future work.

Geographic and temporal limitations. Our research design and technical requirements limited us to measure effects from five countries within a specific time period. Although these countries span multiple regions and account for a substantial share of global web traffic, they cannot fully represent the diversity of the Internet. Further, accurately conducting cross-country measurements introduces practical challenges, as we discussed. It requires physical machines in multiple countries. While we used academic infrastructure in some locations, we relied on personal machines connected to residential networks in others. This setup led to variability in network quality and computational resources, which affected crawl duration and scheduling. Nevertheless, we conducted all crawls within a relatively small time window, between March and May 2026. Despite these challenges, we find sufficient, robust evidence to show that geography is an important factor to be considered in Web measurements.

6.2 Insights and Implications

Insights. Our work highlights that users’ geographical location plays a significant role in shaping the S&P risks they encounter on the web. We observe this effect in two distinct ways. First, the S&P posture of regional websites intended for local users differ from that of globally popular websites. For third-party tracking, *D1* and *D2* sites exhibit similarly high prevalence, while *D3* sites show lower and more variable prevalence across countries. For browser fingerprinting,

the pattern is more monotonic: prevalence is highest in *D1*, lower in *D2*, and lowest in *D3*. This suggests that measurements based only on globally popular websites may not reflect the tracking and fingerprinting behavior users encounter on regional and locally popular websites.

Second, we see geographic variation even when users access the *same set of globally popular websites*. For identical sites in the *D1* catalog, while tracking prevalence changes only modestly across vantage points, the number and set of third-party trackers loaded by the same websites often differ by vantage points. Similarly, fingerprinting varies across vantage points, with WebRTC fingerprinting showing a particularly clear geographic skew.

Our investigation of geographic differences via script inclusion chain analysis also showed divergence, both at first- and third-party script levels. That is, differences in privacy-invasive behaviors, supply-chain or even security-critical inclusion pattern may arise at both first-party and third-party sources, when accessed through different vantage points. Such patterns may result from a range of factors, including location-dependent configuration, region-specific advertising practices, or the use of country/language-driven digital infrastructures. Lastly, we also report inconsistent security-related header deployment patterns by websites across regions and websites, offering differing security controls to users and potentially expose them to myriad of security threats.

Consequently, much of the existent view of the S&P controls on the web are skewed, partial and inaccurate.

Implications. These findings have important implications for multiple stakeholders within the web ecosystem. Web measurement researchers should move beyond globally-popular websites, single vantage point measurements, and VPN/cloud-based infrastructure, as these may mischaracterize the S&P experiences of users who depend on local or regional services and provide a skewed view of the associated risks encountered worldwide. Filter lists maintainers and the ad-blocking community need to expand coverage by incorporating more regional domains and endpoints. Security scanning tools that rely on known heuristics and behavioral signatures may also be affected by missing out capturing critical behavior, and thus need to perform evaluation under more realistic environments. Web developers and service providers should audit region-dependent logic in their code and standardize security configurations to ensure that there are no unintentional variations across regions. Finally, regulators and policymakers should account for geographic deviations in auditing and enforcement, as the same service may exhibit inconsistent S&P controls across jurisdictions.

We argue that future web measurement studies and deployments must treat geographic diversity as a core dimension, despite the practical challenges involved in doing so. Without this, our understanding of web security and privacy will remain incomplete. We hope that our findings motivate future work to broaden the geographic scope of web measurements.

Ethics Considerations

We followed best practices for our Web measurements in line with the established standards of our community [44, 45]. We extensively evaluated the design, methodology, and the impact of our study before we began our data collection step. We detail our assessment in the following:

Crawling Websites & Data Collection: We ensured little to no side effects to the websites we visited during our study as our crawlers only made, at most, ten *stateless* GET requests during our entire study. Additionally, we also did not directly or indirectly interact with the visited website, beyond consenting to the cookie through injecting JavaScript, to limit any potential side effects. Thus, our crawls could not have led to any significant load or state-changing actions on the server, as we observe in practice. Further, we did not feed any fake or erroneous data to any of the websites, nor did we collect any personally-identifiable information at any stage. However, our crawls may violate the Terms of Service of websites that do not allow automated crawling. In this case, we resorted to performing cost-to-benefit analysis and arrived at the conclusion that the findings from our study provide a deeper understanding of the security and privacy postures of websites across geolocations; however, without raising any concerns for the websites or causing harm to individuals.

Crawling Infrastructure: The machines, network, and other infrastructural components used for data collection were either our personal devices or hosted within our university network infrastructure and entirely under our control. As such, we did not share the collected data or Web traffic with any third parties. For our preliminary experiments, we used NordVPN, Windscribe and VyprVPN to identify and understand the differential behavior of websites over VPN and actual headful machines. Here, we do not believe to have leaked any other information to the VPN beyond the list of websites we visited via the network. This does not raise any ethical concerns for the services or individuals, to our knowledge.

Open Science

We plan to open-source the following artifacts associated with our study: the list of CRuX URLs for each country (D1, D2 and D3) we used for our analysis, the *PageGraph* build, the measurement and data collection framework, and the scripts used to analyze the data and formulate results [3]. However, we note that end-to-end reproducibility tests may not yield identical results due to the dynamic nature of the Web. Since the collected data is huge (in TBs), we do not make them public, yet, will share with interested individuals on request.

References

- [1] Gunes Acar, Christian Eubank, Steven Englehardt, Marc Juarez, Arvind Narayanan, and Claudia Diaz. The web never forgets: Persistent tracking mechanisms in the wild. In *ACM CCS*, 2014.
- [2] Abdul Haddi Amjad, Zubair Shafiq, and Muhammad Ali Gulzar. Blocking javascript without breaking the web: An empirical investigation. In *PoPETS*, 2023.
- [3] Anonymous. Artifact repository. <https://anonymous.4open.science/r/global-sec-priv-measurement-C312/README.md>, 2026.
- [4] APNIC. Measuring the Internet. <https://labs.apnic.net/>, 2026. Accessed: 2026-07-18.
- [5] Waqar Aqeel, Balakrishnan Chandrasekaran, Anja Feldmann, and Bruce M. Maggs. On Landing and Internal Web Pages: The Strange Case of Jekyll and Hyde in Web Performance Measurement. In *ACM IMC*, 2020.
- [6] HTTP Archive. Web almanac. <https://almanac.httparchive.org/en/2024/>, 2024.
- [7] HTTP Archive. The web almanac. <https://almanac.httparchive.org/en/2025/>, 2025.
- [8] Pouneh Nikkiah Bahrami, Umar Iqbal, and Zubair Shafiq. Fp-radar: Longitudinal measurement and early detection of browser fingerprinting. In *PoPETS*, 2021.
- [9] Gunjan Balde, Aryendra Singh, Niloy Ganguly, and Mainack Mondal. A Comparative Audit of Privacy Policies from Healthcare Organizations in USA, UK and India. In *arXiv preprint arXiv:2306.11557*, 2023.
- [10] Masudul Hasan Masud Bhuiyan, Matteo Varvello, Cristian-Alexandru Staicu, and Yasir Zaki. Digital Disparities: A Comparative Web Measurement Study Across Economic Boundaries. In *ACM WWW*, 2025.
- [11] Masudul Hasan Masud Bhuiyan, Matteo Varvello, Cristian-Alexandru Staicu, and Yasir Zaki. Non-Western Perspectives on Web Inclusivity: A Study of Accessibility Practices in the Global South. In *IEEE ICSE*, 2025.
- [12] Eleanor Birrell, Jay Rodolitz, Angel Ding, Jenna Lee, Emily McReynolds, Jevan Hutson, and Ada Lerner. SoK: Technical implementation and human impact of internet privacy regulations. In *IEEE S&P*, 2024.
- [13] Christian Böttger, Nurullah Demir, Jan Hörnemann, Bhupendra Acharya, Norbert Pohlmann, Thorsten Holz, Matteo Große-Kampmann, and Tobias Urban. Understanding Regional Filter Lists: Efficacy and Impact. In *PoPETS*, 2025.

- [14] Soumaya Boussaha, Lukas Hock, Miguel Bermejo, Ruben Cuevas Rumin, Angel Cuevas Rumin, David Klein, Martin Johns, Luca Compagna, Daniele Antonoli, and Thomas Barber. Fp-tracer: Fine-grained browser fingerprinting detection via taint-tracking and entropy-based thresholds. In *PoPETS*, 2024.
- [15] Brave Software. Brave pagegraph, 2024. URL <https://github.com/brave/brave-browser/wiki/PageGraph>.
- [16] Brave Software, Inc. Brave. <https://brave.com/>, 2026. URL <https://brave.com/>.
- [17] Stefano Calzavara, Sebastian Roth, Alvis Rabitti, Michael Backes, and Ben Stock. A tale of two headers: a formal analysis of inconsistent Click-Jacking protection on the web. In *USENIX Security*, 2020.
- [18] Chrome Developers. Overview of CrUX, 2025. URL <https://developer.chrome.com/docs/crux>.
- [19] Cloudflare. Domain Categories. <https://developers.cloudflare.com/cloudflare-one/traffic-policies/domain-categories/>, 2026. Accessed: 2026-08-24.
- [20] Cloudflare. Challenges. <https://developers.cloudflare.com/cloudflare-challenges/>, 2026. Accessed: 2026-07-18.
- [21] Cloudflare Radar. Traffic. <https://radar.cloudflare.com/traffic>, 2026. Accessed: 2026-07-18.
- [22] Savino Dambra, Iskander Sanchez-Rola, Leyla Bilge, and Davide Balzarotti. When Sally met trackers: Web tracking from the users’ perspective. In *USENIX Security*, 2022.
- [23] Martin Degeling, Christine Utz, Christopher Lentzsch, Henry Hosseini, Florian Schaub, and Thorsten Holz. We value your privacy... now take some cookies: Measuring the GDPR’s impact on web privacy. In *NDSS*, 2019.
- [24] DuckDuckGo. AutoConsent. <https://github.com/duckduckgo/autoconsent>, 2026. Accessed: 2026-08-24.
- [25] Soumitra Dutta, William H Dutton, and Ginette Law. The new Internet world: A global perspective on freedom of expression, privacy, trust and security online. In *INSEAD*, 2011.
- [26] EasyList. Easylist - overview. <https://easylist.to/>, 2025. Accessed: 2025-08-04.
- [27] EasyList. Other supplementary filter lists and easylist variants, 2025. URL <https://easylist.to/pages/other-supplementary-filter-lists-and-easylist-variants>.
- [28] EasyPrivacy. Easyprivacy - overview. <https://easylist.to/easylist/easyprivacy.txt>, 2025.
- [29] Peter Eckersley. How unique is your web browser? In *International Symposium on Privacy Enhancing Technologies Symposium*. Springer, 2010.
- [30] Steven Englehardt and Arvind Narayanan. Online tracking: A 1-million-site measurement and analysis. In *ACM CCS*, 2016.
- [31] Reza Ghaiumy Anaraky, Yao Li, and Bart Krijnenburg. Difficulties of measuring culture in privacy studies. In *ACM CHI*, 2021.
- [32] Franziska Herbert, Steffen Becker, Leonie Schawewitz, Jonas Hielscher, Marvin Kowalewski, Angela Sasse, Yasemin Acar, and Markus Dürmuth. A world full of privacy and security (mis) conceptions? Findings of a representative survey in 12 countries. In *ACM CHI*, 2023.
- [33] Xuehui Hu, Guillermo Suarez de Tangil, and Nishanth Sastry. Multi-country study of third party trackers from real browser histories. In *IEEE EuroS&P*, 2020.
- [34] Maxime Huyghe, Walter Rudametkin, and Clément Quinton. Fp-rainbow: Fingerprint-based browser configuration identification. In *The Web Conf*, 2025.
- [35] IANA. Root zone database, 2025. URL <https://www.iana.org/domains/root/db>. Accessed: 2025-11-06.
- [36] ICANNWiki. New gtld geographic applications, 2025. URL https://icannwiki.org/New_gTLD_Geographic_Applications. Accessed: 2025-11-06.
- [37] Internet Engineering Task Force. Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile. <https://www.rfc-editor.org/rfc/rfc5280>, 2026. Accessed: 2026-05-02.
- [38] Umar Iqbal, Peter Snyder, Shitong Zhu, Benjamin Livshits, Zhiyun Qian, and Zubair Shafiq. Adgraph: A graph-based approach to ad and tracker blocking. In *IEEE S&P*, 2020.
- [39] Umar Iqbal, Steven Englehardt, and Zubair Shafiq. Fingerprinting the fingerprinters: Learning to detect browser fingerprinting behaviors. In *IEEE S&P*, 2021.
- [40] Nikhil Jha, Martino Trevisan, Luca Vassio, and Marco Mellia. The internet with privacy policies: Measuring the web upon consent. In *ACM TWEB*, 2022.
- [41] Jordan Jueckstock, Shaown Sarker, Peter Snyder, Panagiotis Papadopoulos, Matteo Varvello, Benjamin Livshits, and Alexandros Kapravelos. The blind men and the internet: Multi-vantage point web measurements. In *arXiv preprint arXiv:1905.08767*, 2019.

- [42] Simon Kemp. Digital 2025: Global Overview Report, 2025. URL <https://datareportal.com/reports/digital-2025-global-overview-report>.
- [43] Ken Research. UAE Digital Advertising Market Size, Share & Forecast, By Ad Format, Platform & End-Use Industry, 2026-2032. <https://www.kenresearch.com/industry-reports/uae-digital-advertising-market>, 2026. Accessed: 2026-07-18.
- [44] Erin Kenneally and David Dittrich. The Menlo Report: Ethical Principles Guiding Information and Communication Technology Research. In *Available at SSRN 2445102*, 2012.
- [45] Tadayoshi Kohno, Yasemin Acar, and Wulf Loh. Ethical Frameworks and Computer Security Trolley Problems: Foundations for Conversations. In *USENIX Security*, 2023.
- [46] Latham & Watkins. China’s Cybersecurity Law Amendments Increase Penalties, Broaden Extraterritorial Enforcement. <https://www.lw.com/en/insights/chinas-cybersecurity-law-amendments-increase-penalties-broaden-extraterritorial-enforcement>, 2026. Accessed: 2026-07-18.
- [47] Sebastian Lekies, Ben Stock, Martin Wentzel, and Martin Johns. The Unexpected Dangers of Dynamic {JavaScript}. In *USENIX Security*, 2015.
- [48] Ada Lerner, Anna Kornfeld Simpson, Tadayoshi Kohno, and Franziska Roesner. Internet jones and the raiders of the lost trackers: An archaeological study of web tracking from 1996 to 2016. In *USENIX Security*, 2016.
- [49] Timothy Libert. Exposing the hidden web: An analysis of third-party http requests on 1 million websites. In *arXiv preprint arXiv:1511.00619*, 2015.
- [50] Kyungchan Lim, Yonghwi Kwon, and Doowon Kim. A Longitudinal Study of Vulnerable Client-side Resources and Web Developers’ Updating Behaviors. In *ACM IMC*, 2023.
- [51] Chang Liu and Kirk P Arnett. Raising a red flag on global WWW privacy policies. In *Journal of Computer Information Systems*, 2002.
- [52] Zengrui Liu, Umar Iqbal, and Nitesh Saxena. Opted out, yet tracked: Are regulations enough to protect your privacy? In *PoPETS*, 2022.
- [53] Elisa Luo, Tom Ritter, Stefan Savage, and Geoffrey M Voelker. Canvassing the Fingerprinters: Characterizing Canvas Fingerprinting Use Across the Web. In *ACM IMC*, 2025.
- [54] Jonathan R Mayer and John C Mitchell. Third-party web tracking: Policy and technology. In *IEEE S&P*, 2012.
- [55] Abner Mendoza, Phakpoom Chinprutthiwong, and Guofei Gu. Uncovering HTTP Header Inconsistencies and the Impact on Desktop/Mobile Websites. In *WWW*, 2018.
- [56] Donald R Moscato, Shoshana Altschuller, and Eric D Moscato. Privacy policies on global banks’ websites: does culture matter? In *Communications of the IIMA*, 2013.
- [57] Moti, Zahra and Senol, Asuman and Bostani, Hamid and Borgesius, Frederik Zuiderveen and Moonsamy, Vee-lasha and Mathur, Arunesh and Acar, Gunes. Targeted and troublesome: Tracking and advertising on children’s websites. In *IEEE S&P*, 2024.
- [58] Keaton Mowery and Hovav Shacham. Pixel perfect: Fingerprinting canvas in HTML5. In *W2SP*, 2012.
- [59] Mozilla. Web security cheat sheet, 2025. URL https://infosec.mozilla.org/guidelines/web_security#subresource-integrity.
- [60] Shaoor Munir, Sandra Siby, Umar Iqbal, Steven Englehardt, Zubair Shafiq, and Carmela Troncoso. Cookie-graph: Understanding and detecting first-party tracking cookies. In *ACM CCS*, 2023.
- [61] NordVPN. Nordvpn, 2025. URL <https://nordvpn.com/>. Accessed: 2025-11-06.
- [62] Aysun Ogut, Berke Turanlioglu, Doruk Can Metiner, Albert Levi, Cemal Yilmaz, Orcun Cetin, and Selcuk Uluagac. Dissecting privacy perspectives of websites around the world: Aceptar todo, alle akzeptieren, accept all... In *USENIX Security*, 2024.
- [63] OWASP. Owasp secure headers project, 2025. URL <https://owasp.org/www-project-secure-headers/>.
- [64] Puppeteer. Page.setCacheEnabled() – puppeteer api, 2026. URL <https://pptr.dev/api/puppeteer.page.setCacheEnabled>.
- [65] Audrey Randall, Peter Snyder, Alisha Ukani, Alex C Snoeren, Geoffrey M Voelker, Stefan Savage, and Aaron Schulman. Measuring UID smuggling in the wild. In *ACM IMC*, 2022.
- [66] Sebastian Roth, Michael Backes, and Ben Stock. Assessing the Impact of Script Gadgets on CSP at Scale. In *ACM AsiaCCS*, 2020.

- [67] Sebastian Roth, Timothy Barron, Stefano Calzavara, Nick Nikiforakis, and Ben Stock. Complex security policy? a longitudinal analysis of deployed content security policies. In *NDSS*, 2020.
- [68] Sebastian Roth, Stefano Calzavara, Moritz Wilhelm, Alvise Rabitti, and Ben Stock. The Security Lottery: Measuring Client-Side Web Security Inconsistencies. In *USENIX Security*, 2022.
- [69] Kimberly Ruth, Aureore Fass, Jonathan Azose, Mark Pearson, Emma Thomas, Caitlin Sadowski, and Zakir Durumeric. A world wide view of browsing the world wide web. In *ACM IMC*, 2022.
- [70] Semrush Holdings, Inc. amazon.in website traffic, ranking, analytics (semrush domain overview). <https://www.semrush.com/website/amazon.in/overview/>, 2025. Accessed: 2025-12-13; provides site traffic, ranking and audience data for amazon.in from Semrush’s domain overview tool.
- [71] Tanusree Sharma, Smirity Kaushik, Yaman Yu, Syed Ish-tiaque Ahmed, and Yang Wang. User perceptions and experiences of targeted ads on social media platforms: Learning from bangladesh and india. In *ACM CHI*, 2023.
- [72] Snyder, Peter and Ansari, Lara and Taylor, Cynthia and Kanich, Chris. Browser feature usage on the modern web. In *ACM IMC*, 2016.
- [73] Snyder, Peter and Taylor, Cynthia and Kanich, Chris. Most websites don’t need to vibrate: A cost-benefit approach to improving browser security. In *ACM CCS*, 2017.
- [74] Johnny So, Michael Ferdman, and Nick Nikiforakis. The more things change, the more they stay the same: Integrity of modern javascript. In *The Web Conf*, 2023.
- [75] Brave Software. Brave Shields: Built-in Ad Blocking and Privacy Protection. <https://brave.com/shields/>, 2024. Accessed: 2025-07-27.
- [76] StatCounter. Browser market share worldwide. <https://gs.statcounter.com/browser-market-share>, 2025. Accessed: 2025-08-04.
- [77] Statcounter GlobalStats. Browser Market Share Worldwide. <https://gs.statcounter.com/>, 2026. Accessed: 2026-07-18.
- [78] Marius Steffens, Marius Musch, Martin Johns, and Ben Stock. Who’s hosting the block party? studying third-party blockage of CSP and SRI. In *NDSS*, 2021.
- [79] Ben Stock, Stephan Pfister, Bernd Kaiser, Sebastian Lekies, and Martin Johns. From facepalm to brain bender: Exploring client-side cross-site scripting. In *ACM CCS*, 2015.
- [80] Ben Stock, Martin Johns, Marius Steffens, and Michael Backes. How the Web Tangled Itself: Uncovering the History of {Client-Side} Web ({In} Security). In *USENIX Security*, 2017.
- [81] Rebecca Umbach, Anubha Singh, and Ashley Walker. Your Protection is in Your Hands Only: User Awareness and Adoption of Privacy and Security Practices in Five Majority World Countries. In *Journal of Online Trust and Safety*, 2023.
- [82] Tom Van Goethem, Ping Chen, Nick Nikiforakis, Lieven Desmet, and Wouter Joosen. Large-scale security analysis of the web: Challenges and findings. In *International Conference on Trust and Trustworthy Computing*, 2014.
- [83] Yash Vekaria, Vibhor Agarwal, Pushkal Agarwal, Sangeeta Mahapatra, Sakthi Balan Muthiah, Nishanth Sastry, and Nicolas Kourtellis. Differential tracking across topical webpages of indian news media. In *ACM WebSci*, 2021.
- [84] Yash Vekaria, Yohan Beugin, Shaoor Munir, Gunes Acar, Nataliia Bielova, Steven Englehardt, Umar Iqbal, Alexandros Kapravelos, Pierre Laperdrix, Nick Nikiforakis, et al. Sok: Advances and open problems in web tracking. In *arXiv preprint arXiv:2506.14057*, 2025.
- [85] VyprVPN. Vyprvpn, 2025. URL <https://www.vyprvpn.com/>. Accessed: 2025-11-06.
- [86] W3Schools. HTML Standard: Links. https://www.w3schools.com/html/html_favicon.asp, 2026. Accessed: 2026-05-02.
- [87] Lukas Weichselbaum, Michele Spagnuolo, Sebastian Lekies, and Artur Janc. CSP is Dead, Long Live CSP! On the Insecurity of Whitelists and the Future of Content Security Policy. In *ACM CCS*, 2016.
- [88] Zachary Weinberg, Diogo Barradas, and Nicolas Christin. Chinese wall or Swiss cheese? keyword filtering in the Great Firewall of China. In *WWW*, 2021.
- [89] Windscribe. Windscribe vpn, 2025. URL <https://windscribe.com/>. Accessed: 2025-11-06.
- [90] Diwen Xue, Benjamin Mixon-Baca, ValdikSS, Anna Ablove, Beau Kujath, Jedidiah R Crandall, and Roya Ensafi. Tspu: Russia’s decentralized censorship system. In *ACM IMC*, 2022.

- [91] Yandex. GDPR compliance. <https://yandex.com/support/metrica/en/general/gdpr>, 2026. Accessed: 2026-07-18.
- [92] Zhonghao Yu, Sam Macbeth, Konark Modi, and Josep M Pujol. Tracking the trackers. In *WWW*, 2016.
- [93] Luqman Zagi, Zahra Moti, and Gunes Acar. Referrer Policy: Implementation and Circumvention. In *PoPETS*, 2025.
- [94] Sebastian Zimmeck and Steven M Bellovin. Privee: An architecture for automatically analyzing web privacy policies. In *USENIX Security*, 2014.
- [95] Sebastian Zimmeck, Oliver Wang, Kuba Alicki, Jocelyn Wang, and Sophie Eng. Usability and enforceability of global privacy control. In *PoPETS*, 2023.

A Construction and Validation of *D2* Catalog

The *D2* catalog contains country-coded versions of global websites. For example, `example.dz` is considered for Algeria if a corresponding global version, such as `example.com` or `example.org`, appears in the global CRuX list. We follow a two-stage approach to build the *D2* catalog.

We first collect candidate *D2* sites by comparing the top 1 million sites for each country with the top 1 million global sites from CRuX. For each country, we retain a site as a *D2* candidate if it satisfies three conditions: (1) it does not appear in the *D1* catalog, (2) it uses one of the country’s TLDs, and (3) its second-level domain appears in the global CRuX list under a different public suffix. This step produced 239 candidates for Algeria, 7,684 for Germany, 6,490 for India, 1,277 for the UAE, and 1,045 for the USA. We then validate these candidates using two signals: favicon similarity and TLS certificates.

A.1 Favicon Similarity Validation

Given that favicons provide a visual identifier of a website’s brand, we assume that when a country-coded candidate and its global equivalent use a visually comparable favicon, the two domains likely belong to the same brand.

We attempt to collect favicons for each candidate *D2* site and its global equivalent by parsing `<link rel="icon">` tags in the page HTML. We successfully collected favicons for the majority of candidate sites across all countries: 70.6% for Algeria, 74.6% for India, 82.1% for the UAE, 82.4% for the USA, and 90.4% for Germany. We also collected favicons for 82.5% of their global equivalents. To measure visual similarity, we first normalize all favicons so that differences in transparency, aspect ratio, and size do not affect the comparison. We render them on a white background, pad them to square images while preserving their aspect ratios, and resize

them to 224×224 pixels. We then use a CLIP ViT-B/32 image encoder to represent each normalized favicon as a feature embedding and compute the cosine similarity between the embeddings of each candidate favicon and its global equivalent. A higher cosine similarity indicates that the two favicons are visually more likely to be similar.

We consider a candidate *D2* site to be *favicon-validated* if the cosine similarity with its global equivalent is greater than or equal to 0.85. We tested several threshold values: 0.70, 0.75, 0.80, 0.85, 0.90, and 0.95. For each threshold, we manually inspected 50 comparison results. The threshold of 0.85 achieved a 100% success rate in this manual inspection. Using favicon similarity, we validated 24.7% of candidates for Algeria, 54.3% for Germany, 23.2% for India, 36.5% for the UAE, and 34.6% for the USA.

A.2 TLS Certificate Validation

TLS certificates provide a complementary signal of shared ownership. Websites that belong to the same organization, or that are managed as part of the same service, may use certificates that cover multiple related domains or that are issued to the same legal entity. We use this information to identify candidate *D2* sites operated by corresponding global equivalents. We collect TLS certificates for all candidate *D2* sites and their mapped global equivalents, and extract two fields from each certificate: Subject Alternative Names (SANs) and *organizationName*. SANs list the domain names explicitly covered by a certificate, while the *organizationName* field identifies the legal entity to which the certificate was issued. We retrieved TLS data for over 92% of candidate sites across all countries: 95.5% for Algeria, 96.4% for Germany, 92.7% for India, 96.0% for the UAE, and 96.5% for the USA.

We consider a candidate *D2* site to be *TLS-validated* if at least one of the following conditions holds for the candidate and its global equivalent: (1) the equivalent’s hostname is covered by one of the candidate’s SANs; (2) the candidate’s hostname is covered by one of the equivalent’s SANs; (3) the two sites share an identical SAN set; or (4) both sites have a non-empty and matching *organizationName* value.

TLS validation rates were lower than favicon validation rates: 6.3% for Algeria, 8.1% for Germany, 2.3% for India, 8.6% for the UAE, and 1.8% for the USA. Overall, combining both validation signals, we constructed final *D2* catalogs containing 65 sites for Algeria, 4,278 for Germany, 1,535 for India, 505 for the UAE, and 371 for the USA. These catalogs could be expanded through using adjusting the similarity thresholds or additional validation signals. However, we resorted to a high-confidence catalog of country-coded global websites for this study.

B Additional Tables & Figures

Table 4: Country-specific eTLDs extracted from IANA and ICANN database that we considered for website cataloging.

Country	eTLDs
Algeria	.dz
Germany	.de, .bayern, .berlin, .hamburg, .nrw, .saarland, .cologne, .koeln
India	.in
UAE	.ae, .abudhabi, .dubai
USA	.us, .boston, .miami, .nyc

Table 5: Security-related HTTP Response Headers

Access-Control-Allow-Credentials, Access-Control-Allow-Headers, Access-Control-Allow-Methods, Access-Control-Allow-Origin, Access-Control-Expose-Headers, Access-Control-Max-Age, Content-Security-Policy, Content-Security-Policy-Report-Only, Cross-Origin-Embedder-Policy, Cross-Origin-Opener-Policy, Cross-Origin-Resource-Policy, Expect-CT, Origin-Agent-Cluster, Permissions-Policy, Reporting-Endpoints, Report-To, Strict-Transport-Security, Timing-Allow-Origin, X-Content-Type-Options, X-Frame-Options, X-Permitted-Cross-Domain-Policies, X-XSS-Protection

Figure 2: Prevalence (gain/loss count) of Top 20 third-party script sources on websites when accessed over VPN.

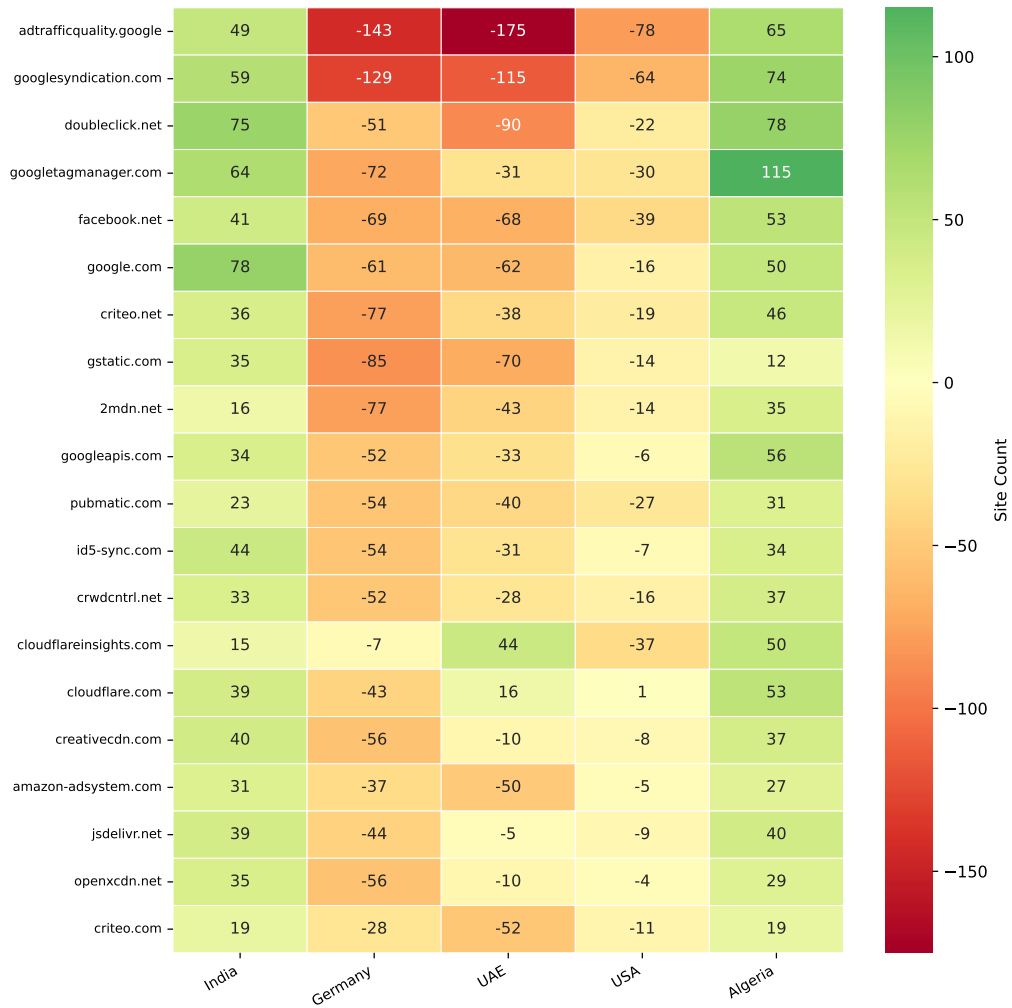


Figure 3: Distribution (gain/loss %) of broad classes of JavaScript API invocations on websites when accessed over VPN.

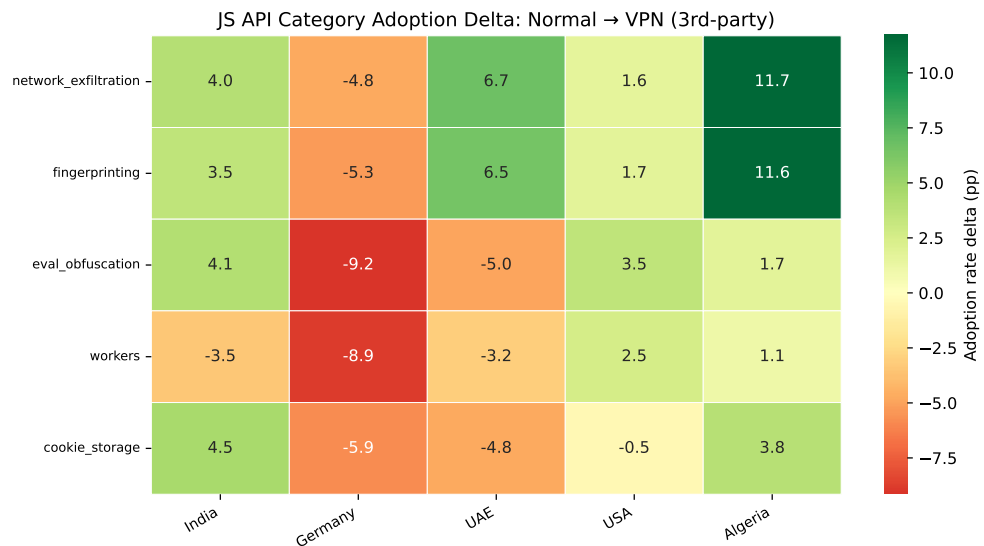


Figure 4: Prevalence (gain/loss count) of Top 20 third-party iframe sources on websites when accessed over VPN.

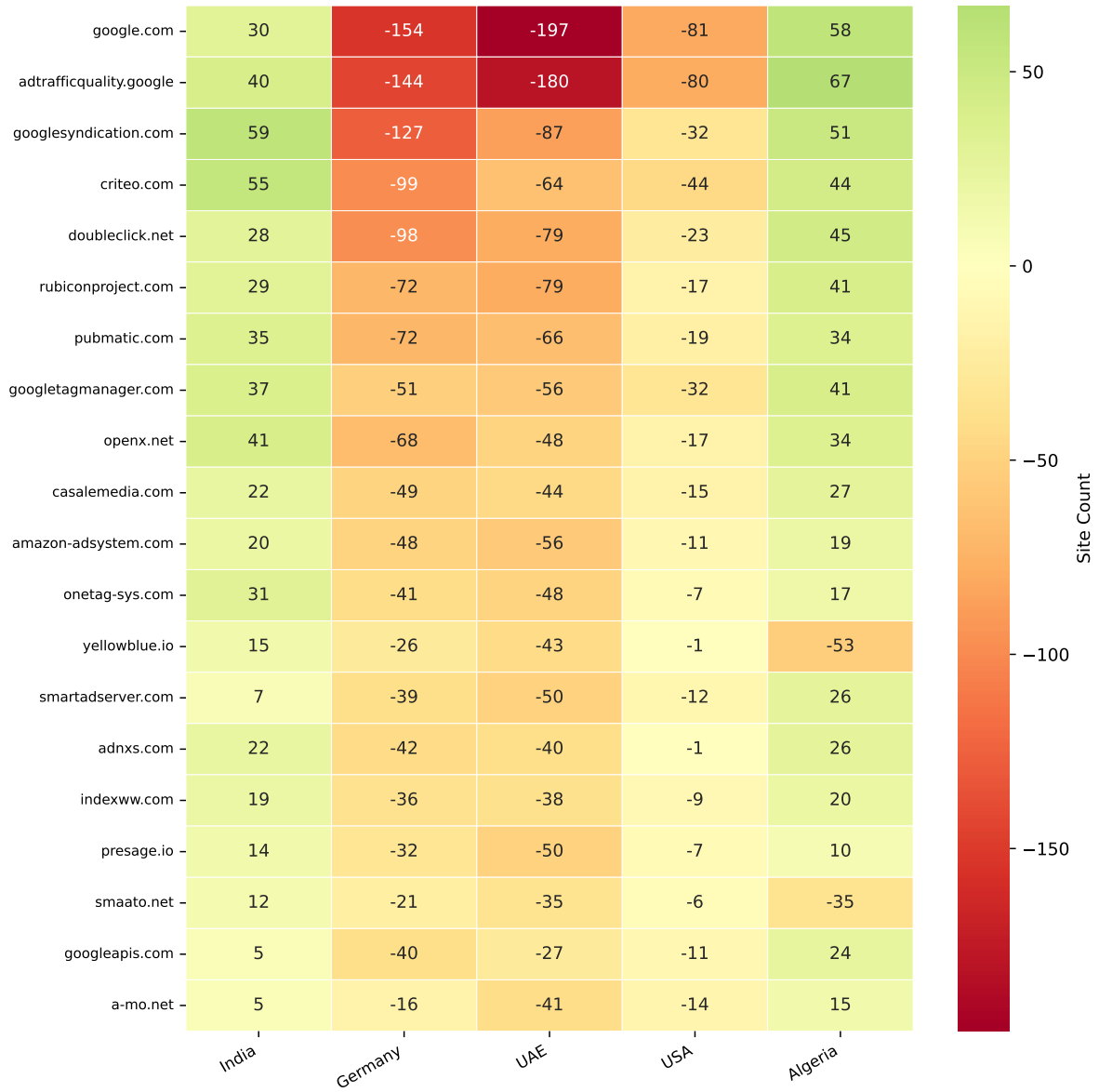


Table 6: Header Deployment over VPN: Comparison of deployment of security-related HTTP headers on Top 1,500 globally popular websites (D1) when accessed with and without VPN. (Note that we could not collect VPN data from Algeria due to frequent network and electricity outages, and hence, we do not report them as of time of submission.)

HTTP Headers	GERMANY (VPN / Physical)	INDIA (VPN / Physical)	UAE (VPN / Physical)	USA (VPN / Physical)
Set-Cookie	43.49% / 46.70%	45.10% / 44.83%	43.23% / 45.99%	43.58% / 47.77%
Strict-Transport-Security	42.16% / 44.56%	42.16% / 40.11%	40.64% / 42.60%	41.62% / 44.74%
X-Content-Type-Options	38.32% / 40.11%	38.68% / 37.43%	37.34% / 40.29%	38.50% / 40.82%
X-Frame-Options	38.15% / 39.66%	38.24% / 36.27%	36.99% / 38.41%	37.97% / 39.84%
X-XSS-Protection	28.61% / 29.77%	29.14% / 28.07%	28.61% / 30.04%	28.88% / 30.21%
Report-To	26.65% / 27.45%	26.92% / 26.92%	25.76% / 26.47%	26.65% / 27.36%
Content-Security-Policy	23.53% / 25.04%	23.89% / 23.35%	22.91% / 24.60%	23.44% / 25.40%
Access-Control-Allow-Origin	9.63% / 9.63%	9.00% / 8.47%	9.27% / 9.09%	9.54% / 9.54%
Permissions-Policy	6.77% / 7.04%	6.86% / 7.04%	6.60% / 7.04%	7.31% / 8.20%
Cross-Origin-Opener-Policy	4.37% / 4.63%	4.37% / 4.19%	3.92% / 5.17%	4.46% / 4.63%
Access-Control-Allow-Headers	4.28% / 4.55%	4.37% / 4.10%	3.92% / 4.37%	4.46% / 4.72%
Access-Control-Allow-Methods	4.19% / 4.55%	4.19% / 3.83%	3.65% / 4.10%	4.37% / 4.46%
X-Permitted-Cross-Domain-Policies	4.10% / 4.01%	4.10% / 4.10%	4.10% / 4.01%	3.65% / 4.37%
Content-Security-Policy-Report-Only	3.12% / 3.21%	3.12% / 2.94%	2.67% / 3.65%	3.21% / 3.74%
Access-Control-Allow-Credentials	2.94% / 2.94%	2.94% / 2.67%	2.41% / 3.03%	3.12% / 3.30%
Cross-Origin-Resource-Policy	2.50% / 2.67%	2.41% / 2.50%	2.41% / 2.94%	2.50% / 2.85%
Reporting-Endpoints	1.87% / 2.05%	2.23% / 2.14%	2.05% / 2.32%	2.05% / 2.41%
Access-Control-Expose-Headers	2.50% / 2.41%	2.23% / 1.87%	2.14% / 1.78%	2.14% / 2.41%
Origin-Agent-Cluster	1.52% / 1.69%	1.69% / 1.52%	1.52% / 1.78%	1.60% / 1.78%
Expect-CT	1.52% / 1.60%	1.25% / 1.34%	1.52% / 1.34%	1.25% / 1.43%
Access-Control-Max-Age	1.25% / 1.34%	1.34% / 0.98%	0.89% / 1.34%	1.25% / 1.43%
Timing-Allow-Origin	0.53% / 0.71%	0.62% / 0.71%	0.62% / 0.62%	0.62% / 0.71%
Cross-Origin-Embedder-Policy	0.62% / 0.62%	0.36% / 0.36%	0.36% / 0.62%	0.36% / 0.36%

Table 7: Overall Distributions of security-related header deployments across different countries and catalog of websites (D1, D2 and D3). Please note that the distribution is computed where the denominator is the number of eTLDs for each country where at least one request was successful (e.g., response status code was 200)

HTTP Headers	URL Categories	ALGERIA	GERMANY	INDIA	UAE	USA	Mean	Std. Dev
X-Content-Type-Options	D1	31.83%	33.27%	29.04%	33.89%	19.33%	29.47%	5.97%
	D2	41.82%	44.59%	31.44%	47.56%	19.68%	37.02%	11.44%
	D3	29.47%	47.45%	39.18%	29.87%	39.28%	37.05%	7.53%
Strict-Transport-Security	D1	38.93%	40.71%	34.47%	41.14%	24.02%	35.85%	7.12%
	D2	41.82%	47.25%	31.35%	53.58%	28.72%	40.54%	10.50%
	D3	19.79%	45.63%	36.00%	32.51%	37.58%	34.30%	9.43%
X-Frame-Options	D1	32.21%	34.19%	28.76%	34.40%	19.64%	29.84%	6.13%
	D2	32.73%	41.68%	27.04%	45.56%	15.96%	32.59%	11.81%
	D3	27.13%	40.81%	40.54%	28.10%	33.22%	33.96%	6.55%
X-XSS-Protection	D1	23.15%	24.16%	20.96%	25.07%	14.00%	21.47%	4.45%
	D2	25.45%	29.46%	20.35%	32.95%	12.77%	24.20%	7.93%
	D3	21.77%	27.51%	33.11%	19.40%	20.58%	24.48%	5.74%
Content-Security-Policy	D1	20.49%	21.71%	18.81%	22.11%	12.69%	19.16%	3.84%
	D2	16.36%	22.56%	23.83%	29.80%	16.49%	21.81%	5.62%
	D3	8.41%	23.14%	25.24%	23.99%	26.76%	21.51%	7.45%
Report-To	D1	20.52%	21.09%	16.79%	20.52%	13.49%	18.48%	3.27%
	D2	1.82%	16.01%	16.32%	19.77%	7.45%	12.27%	7.40%
	D3	1.73%	7.12%	8.41%	12.85%	10.03%	8.03%	4.12%
Permissions-Policy	D1	7.21%	7.16%	6.37%	7.79%	4.74%	6.65%	1.18%
	D2	1.82%	7.25%	7.24%	7.45%	12.23%	7.20%	3.69%
	D3	6.63%	6.68%	13.01%	8.01%	5.03%	7.87%	3.06%
Access-Control-Allow-Origin	D1	10.31%	11.23%	9.43%	11.58%	6.20%	9.75%	2.15%
	D2	7.27%	9.39%	8.16%	11.46%	6.91%	8.64%	1.84%
	D3	5.00%	6.83%	11.67%	7.08%	5.43%	7.20%	2.65%
X-Permitted-Cross-Domain-Policies	D1	2.47%	2.74%	2.24%	2.91%	1.42%	2.36%	0.58%
	D2	1.82%	6.09%	3.67%	8.60%	4.26%	4.89%	2.57%
	D3	3.93%	4.66%	6.40%	4.60%	5.96%	5.11%	1.03%
Cross-Origin-Resource-Policy	D1	1.71%	1.70%	1.65%	1.65%	0.92%	1.53%	0.34%
	D2	0.00%	0.88%	0.92%	0.29%	0.53%	0.52%	0.39%
	D3	1.17%	0.55%	1.26%	1.06%	0.77%	0.96%	0.30%
Content-Security-Policy-Report-Only	D1	2.38%	2.48%	2.29%	2.38%	1.41%	2.19%	0.44%
	D2	0.00%	3.29%	1.19%	3.15%	2.13%	1.95%	1.38%
	D3	0.20%	1.25%	0.65%	0.55%	0.85%	0.70%	0.39%
Access-Control-Allow-Methods	D1	4.15%	4.23%	3.75%	4.37%	2.41%	3.78%	0.80%
	D2	3.64%	2.35%	3.21%	2.87%	1.60%	2.73%	0.79%
	D3	1.68%	2.61%	6.77%	3.13%	2.02%	3.24%	2.05%
Cross-Origin-Opener-Policy	D1	3.02%	3.07%	2.57%	2.96%	1.39%	2.60%	0.71%
	D2	0.00%	1.61%	1.10%	0.57%	1.06%	0.87%	0.61%
	D3	1.99%	1.20%	1.49%	1.42%	1.34%	1.49%	0.30%
Access-Control-Allow-Credentials	D1	3.04%	3.36%	2.71%	3.49%	2.06%	2.93%	0.57%
	D2	3.64%	1.44%	1.65%	1.15%	2.13%	2.00%	0.98%
	D3	2.65%	1.84%	3.26%	1.07%	0.54%	1.87%	1.11%
Reporting-Endpoints	D1	1.31%	1.30%	1.23%	1.42%	0.85%	1.22%	0.22%
	D2	0.00%	1.12%	0.64%	1.15%	0.00%	0.58%	0.57%
	D3	0.05%	0.54%	0.18%	0.41%	1.39%	0.51%	0.53%
Access-Control-Allow-Headers	D1	4.12%	4.33%	3.65%	4.53%	2.43%	3.81%	0.84%
	D2	0.00%	2.45%	3.02%	2.87%	2.66%	2.20%	1.25%
	D3	3.93%	2.74%	6.44%	3.15%	2.09%	3.67%	1.69%
Cross-Origin-Embedder-Policy	D1	0.71%	0.60%	0.59%	0.63%	0.39%	0.58%	0.12%
	D2	0.00%	0.35%	0.46%	0.00%	0.53%	0.27%	0.25%
	D3	1.07%	0.36%	0.89%	0.92%	0.33%	0.71%	0.34%
Expect-CT	D1	1.04%	1.26%	0.94%	1.11%	0.75%	1.02%	0.19%
	D2	1.82%	0.84%	1.37%	2.29%	1.60%	1.58%	0.54%
	D3	0.76%	0.91%	3.96%	1.17%	0.96%	1.55%	1.35%
Access-Control-Expose-Headers	D1	1.75%	1.86%	1.93%	2.07%	1.12%	1.75%	0.37%
	D2	0.00%	2.84%	2.47%	3.15%	2.66%	2.22%	1.27%
	D3	0.20%	0.99%	1.31%	1.30%	1.97%	1.15%	0.64%
Timing-Allow-Origin	D1	1.11%	1.26%	0.99%	1.46%	0.79%	1.12%	0.25%
	D2	0.00%	2.07%	1.01%	2.29%	2.13%	1.50%	0.98%
	D3	0.00%	0.81%	0.82%	1.07%	1.60%	0.86%	0.58%
Access-Control-Max-Age	D1	1.47%	1.40%	1.37%	1.47%	0.94%	1.33%	0.22%
	D2	0.00%	0.63%	0.64%	0.86%	0.53%	0.53%	0.32%
	D3	0.05%	0.37%	1.26%	0.30%	0.26%	0.45%	0.47%
Origin-Agent-Cluster	D1	0.67%	0.83%	0.69%	0.76%	0.42%	0.67%	0.16%
	D2	0.00%	0.32%	0.46%	0.29%	0.00%	0.21%	0.20%
	D3	0.20%	0.10%	0.09%	0.14%	0.31%	0.17%	0.09%

Table 8: Effective policy enforcements of CSP header deployments

CSP Flags	URL Categories	ALGERIA	GERMANY	INDIA	UAE	USA	Mean	Std. Dev
Prone to XSS	D1	84.49%	85.92%	82.27%	86.02%	82.97%	84.33%	1.70%
	D2	91.67%	92.65%	94.48%	94.40%	91.67%	92.97%	1.40%
	D3	89.76%	88.01%	95.38%	97.08%	81.75%	90.40%	6.13%
Prone to TLS attacks	D1	71.21%	72.28%	73.11%	71.31%	75.60%	72.70%	1.80%
	D2	75.00%	72.24%	46.51%	68.00%	33.33%	59.02%	18.21%
	D3	76.10%	81.45%	65.94%	42.13%	68.88%	66.90%	15.12%
Prone to framing attacks	D1	39.81%	38.55%	40.30%	39.60%	37.75%	39.20%	1.03%
	D2	50.00%	31.20%	58.43%	34.40%	22.22%	39.25%	14.68%
	D3	50.24%	44.84%	56.10%	67.12%	35.03%	50.67%	12.03%
Invalid Policy	D1	5.92%	5.85%	6.19%	5.68%	6.08%	5.94%	0.20%
	D2	8.33%	11.51%	7.27%	10.40%	13.89%	10.28%	2.62%
	D3	5.85%	7.35%	8.62%	6.03%	5.09%	6.59%	1.40%

Table 9: Effective policy enforcements of HSTS header deployments

HSTS Flags	URL Categories	ALGERIA	GERMANY	INDIA	UAE	USA	Mean	Std. Dev
Does not include <i>preload</i>	D1	73.75%	73.11%	72.19%	73.10%	73.25%	73.08%	0.56%
	D2	90.32%	75.41%	71.19%	71.96%	70.97%	75.97%	8.22%
	D3	55.29%	81.09%	61.00%	73.09%	83.73%	70.84%	12.40%
Missing <i>includeSubDomains</i>	D1	39.23%	37.88%	39.24%	38.59%	39.43%	38.87%	0.64%
	D2	54.84%	43.00%	37.86%	42.06%	54.84%	46.52%	7.84%
	D3	27.86%	41.58%	25.01%	35.78%	47.13%	35.47%	9.23%
Non-standard <i>max-age</i>	D1	24.87%	24.71%	24.33%	24.86%	23.57%	24.47%	0.55%
	D2	38.71%	22.75%	22.86%	24.77%	11.29%	24.08%	9.76%
	D3	19.65%	19.63%	13.70%	21.69%	16.69%	18.27%	3.12%
Disabled	D1	1.60%	1.63%	1.82%	1.44%	1.62%	1.62%	0.13%
	D2	0.00%	0.55%	0.00%	1.40%	0.00%	0.39%	0.61%
	D3	0.22%	1.20%	2.19%	2.67%	3.54%	1.96%	1.29%

Table 10: Effective policy enforcements of X-FO header deployments

X-FO Flags	URL Categories	ALGERIA	GERMANY	INDIA	UAE	USA	Mean	Std. Dev
SELF	D1	78.40%	78.14%	77.86%	78.92%	78.75%	78.41%	0.43%
	D2	88.00%	79.84%	80.62%	86.10%	78.95%	82.70%	4.07%
	D3	77.35%	73.77%	71.55%	82.48%	87.56%	78.54%	6.52%
NONE	D1	19.03%	19.36%	20.12%	18.48%	19.00%	19.19%	0.61%
	D2	8.00%	16.57%	15.50%	11.23%	18.42%	13.94%	4.24%
	D3	14.48%	23.22%	26.37%	16.01%	11.21%	18.26%	6.31%
UNSAFE	D1	2.57%	2.50%	2.02%	2.60%	2.25%	2.39%	0.25%
	D2	4.00%	3.59%	3.88%	2.67%	2.63%	3.35%	0.66%
	D3	8.17%	3.01%	2.07%	1.51%	1.23%	3.20%	2.86%

Table 11: Effective policy enforcements of Set-Cookie header deployments. Note: Since *PageGraph* only captures cookies visible at JS level, we do not report *httpOnly*-related statistics here.

Set-Cookie Attributes	URL Categories	ALGERIA	GERMANY	INDIA	UAE	USA	Mean	Std. Dev
SameSite: Strict	D1	0.16%	0.12%	0.19%	0.15%	0.20%	0.16%	0.03%
	D2	0.00%	0.21%	0.63%	1.52%	0.00%	0.47%	0.64%
	D3	0.08%	0.07%	0.10%	0.16%	0.15%	0.11%	0.04%
SameSite: Lax	D1	0.25%	0.24%	0.31%	0.28%	0.29%	0.27%	0.03%
	D2	0.80%	0.81%	0.04%	0.14%	0.62%	0.48%	0.36%
	D3	0.07%	0.87%	0.37%	0.09%	0.59%	0.40%	0.34%
SameSite: None	D1	99.59%	99.64%	99.50%	99.57%	99.51%	99.56%	0.06%
	D2	99.20%	98.98%	99.33%	98.34%	99.38%	99.05%	0.43%
	D3	99.85%	99.05%	99.52%	99.75%	99.27%	99.49%	0.33%
<i>secure</i> set to <i>false</i>	D1	99.69%	99.49%	99.14%	99.52%	99.39%	99.45%	0.20%
	D2	99.99%	99.00%	99.58%	99.38%	99.22%	99.43%	0.38%
	D3	99.83%	99.19%	99.53%	99.25%	99.74%	99.51%	0.29%

Table 12: Distribution of *third-party* scripts included by websites across countries and URL catalogs (*D1*, *D2*, and *D3*). The script origins are ordered in descending order of their occurrences across the cumulative sum of websites and across catalogs.

Script Sources	URL Catalogs	ALGERIA	GERMANY	INDIA	UAE	USA	Mean	Std. Dev
www.googletagmanager.com	D1	59.37%	56.4%	48.87%	59.39%	32.74%	51.35%	11.25%
	D2	56.36%	63.44%	56.4%	71%	28.28%	55.10%	16.16%
	D3	25.54%	33.95%	32.21%	62.18%	49.13%	40.60%	14.83%
googleads.g.doubleclick.net	D1	30.5%	24.62%	22.03%	33.31%	15.38%	25.17%	7.08%
	D2	18.18%	20.65%	24.98%	42.01%	11.11%	23.38%	11.56%
	D3	6.12%	12.81%	9.88%	34.46%	11.32%	14.92%	11.21%
www.google.com	D1	26.48%	26.03%	18.94%	27.93%	13.98%	22.67%	5.98%
	D2	29.09%	19.95%	19.96%	36.04%	14.65%	23.94%	8.53%
	D3	15.7%	10.99%	13.37%	28.11%	27.58%	19.15%	8.11%
connect.facebook.net	D1	20.35%	20.08%	15.43%	21.83%	10.75%	17.69%	4.56%
	D2	38.18%	27.65%	22.56%	41.73%	11.62%	28.35%	12.14%
	D3	9.48%	11.23%	8.54%	23.72%	10.79%	12.75%	6.22%
maps.googleapis.com	D1	12.37%	13.84%	8.7%	14.83%	7.59%	11.47%	3.18%
	D2	16.36%	9.11%	15.04%	17.34%	7.07%	12.98%	4.60%
	D3	18.86%	7.48%	14.5%	27.24%	25.68%	18.75%	8.14%
www.google-analytics.com	D1	17.11%	17.09%	14.1%	18.54%	10.08%	15.38%	3.38%
	D2	16.36%	12.14%	17.73%	19.51%	7.58%	14.66%	4.81%
	D3	11.98%	7.95%	11.1%	16.97%	18.66%	13.33%	4.40%
cdnjs.cloudflare.com	D1	10.73%	22.04%	17.58%	18.42%	14.39%	16.63%	4.28%
	D2	7.27%	12.44%	13.52%	19.24%	7.58%	12.01%	4.92%
	D3	7.34%	4.79%	11.65%	12.8%	12.07%	9.73%	3.49%
static.cloudflareinsights.com	D1	23.63%	12.01%	19.95%	23.02%	13.51%	18.42%	5.38%
	D2	0%	5.5%	17.99%	16.26%	10.1%	9.97%	7.47%
	D3	2.34%	2.78%	8.71%	12.15%	15%	8.20%	5.61%
www.gstatic.com	D1	10.93%	12.75%	7.53%	13.07%	6.32%	10.12%	3.06%
	D2	18.18%	14.11%	13.07%	24.93%	9.09%	15.88%	6.01%
	D3	19.47%	8.33%	11.4%	32.58%	15.37%	17.43%	9.45%
pagead2.googlesyndication.com	D1	20.86%	22.18%	13.42%	24.96%	11.76%	18.64%	5.74%
	D2	7.27%	4.94%	10.47%	9.49%	4.55%	7.34%	2.65%
	D3	2.45%	7.21%	7.71%	2.21%	2.83%	4.48%	2.73%
ep2.adtrafficquality.google	D1	17.07%	19.75%	9.6%	21.82%	10.29%	15.71%	5.53%
	D2	7.27%	4.4%	10.03%	7.59%	4.04%	6.67%	2.48%
	D3	2.34%	7%	6.97%	2.17%	2.68%	4.23%	2.52%
cdn.jsdelivr.net	D1	9.49%	10.71%	6.69%	11.33%	5.32%	8.71%	2.60%
	D2	3.64%	7.84%	8.5%	11.92%	4.04%	7.19%	3.43%
	D3	8%	3.52%	10.13%	11.92%	5.98%	7.91%	3.32%
www.clarity.ms	D1	8.91%	8.83%	6.6%	9.35%	5.08%	7.75%	1.84%
	D2	3.64%	12.74%	8.42%	16.8%	8.08%	9.94%	5.01%
	D3	0.51%	4.14%	4.49%	10.73%	2.85%	4.54%	3.79%
bat.bing.com	D1	6.31%	5.94%	4.22%	6.77%	3%	5.25%	1.59%
	D2	3.64%	17.24%	4.39%	12.2%	6.57%	8.81%	5.79%
	D3	0%	5.7%	0.82%	0.97%	1.62%	1.82%	2.24%
static.criteo.net	D1	8.03%	10.64%	4.41%	11.8%	4.76%	7.93%	3.34%
	D2	0%	2.03%	0.98%	2.17%	1.01%	1.24%	0.89%
	D3	0%	3%	0.61%	0.13%	0.24%	0.79%	1.25%
invstatic101.creativecdn.com	D1	8.23%	9.69%	4.41%	10.75%	4.7%	7.56%	2.88%
	D2	0%	2.1%	0.9%	2.98%	1.01%	1.40%	1.16%
	D3	0%	1.63%	0.24%	0.13%	0.13%	0.43%	0.68%
code.jquery.com	D1	3.34%	3.58%	2.84%	3.9%	1.96%	3.13%	0.76%
	D2	1.82%	3.37%	5.55%	6.23%	2.02%	3.80%	2.02%
	D3	3.72%	1.7%	7.83%	6.24%	5.59%	5.02%	2.37%
cdn.id5-sync.com	D1	8.25%	8.63%	4.55%	10.22%	4.12%	7.15%	2.68%
	D2	0%	1.2%	0.81%	0.54%	1.52%	0.81%	0.59%
	D3	0%	3.18%	0.45%	0.04%	0.24%	0.78%	1.35%
c.amazon-adsystem.com	D1	7.98%	8.49%	5.26%	9.26%	4.07%	7.01%	2.23%
	D2	0%	2.13%	1.34%	1.63%	1.01%	1.22%	0.80%
	D3	0%	2.74%	0.27%	0.15%	0.34%	0.70%	1.15%
s0.2mdn.net	D1	5.41%	8.59%	2.13%	11.92%	4.55%	6.52%	3.80%
	D2	0%	0.7%	0.9%	1.9%	1.01%	0.90%	0.68%
	D3	0%	2.89%	0.34%	0.34%	0.52%	0.82%	1.17%

Table 13: Distribution of 3P-script *request volume* loaded by websites across countries and URL catalogs (*D1*, *D2*, and *D3*).

Metric	URL Catalogs	ALGERIA	GERMANY	INDIA	UAE	USA	Mean	Std. Dev
Median 3P-scripts per site	D1	105	97	42	132	39	83.00	40.92
	D2	80.5	138	69	240	87	122.90	70.59
	D3	45	66	27	115	43	59.20	34.14
Sites with ≥ 1 3P-scripts (%)	D1	92.54%	92.84%	92.46%	92.81%	92.93%	92.72%	0.20%
	D2	86.79%	93.17%	90.54%	96.10%	79.55%	89.23%	6.40%
	D3	66.24%	69.94%	67.43%	88.47%	86.33%	75.68%	10.81%
3P share of all script requests (%)	D1	73.61%	67.19%	59.59%	68.66%	63.28%	66.47%	5.33%
	D2	71.07%	49.83%	67.65%	58.91%	59.43%	61.38%	8.31%
	D3	42.87%	45.46%	48.14%	49.98%	55.01%	48.29%	4.62%

Table 14: Top 10 third-party tracking domains across the combined D1/D2/D3 catalogs. Values are the percentage of sites on which each tracker was observed.

Tracker	ALGERIA	GERMANY	INDIA	UAE	USA
googletagmanager.com	42.3%	45.4%	37.2%	53.0%	47.8%
google-analytics.com	29.7%	28.0%	27.5%	37.8%	35.2%
doubleclick.net	27.4%	27.9%	21.8%	37.0%	25.6%
google.com	25.0%	26.8%	21.0%	35.5%	23.7%
googlesyndication.com	17.2%	20.8%	12.4%	12.4%	12.6%
cloudflareinsights.com	15.0%	6.2%	12.5%	14.7%	16.5%
facebook.net	11.1%	14.2%	9.5%	17.9%	10.4%
facebook.com	10.4%	13.6%	8.7%	17.5%	9.7%
googleapis.com	7.5%	5.6%	5.1%	13.7%	6.7%
adnxs.com	9.5%	8.6%	3.6%	7.9%	6.8%

Table 15: Trackers observed only from a specific vantage point. Each entry reports the percentage of sites in that country on which the tracker was observed.

ALGERIA	GERMANY	INDIA	UAE	USA
google.dz (17.5%)	google.de (16.8%)	google.co.in (13.7%)	google.ae (20.8%)	tpmn.co.kr (0.82%)
51.la (0.13%)	sparkasse.de (1.01%)	stripchatgirls.com (0.37%)	superchat.live (0.48%)	monsido.com (0.15%)
adventori.com (0.13%)	ad-srv.net (0.64%)	adster.tech (0.16%)	tildacdn.one (0.28%)	nublet.shop (0.13%)
fawnlux.com (0.11%)	ioam.de (0.46%)	gokwik.co (0.08%)	2d86bbe43f.com (0.09%)	b3953dfb4c.com (0.10%)
naltis.com (0.10%)	hyj.mobi (0.30%)	stat.ovh (0.08%)	pippio.com (0.07%)	thtrck.com (0.09%)

Table 16: Country-level prevalence of third-party tracking and fingerprinting in *D3*. Values report 95% confidence intervals estimated using non-parametric bootstrap resampling.

Country	3P-tracking prevalence CI	Fingerprinting prevalence CI
Algeria	[33.21, 36.81]	[0.27, 0.83]
Germany	[49.64, 51.86]	[5.07, 6.08]
India	[35.20, 37.25]	[1.90, 2.54]
UAE	[64.02, 66.04]	[5.83, 6.86]
USA	[54.26, 56.63]	[7.45, 8.75]